

Feature catalogue printout of S125

Version: 0.0.4 with FIHO_FIXES 2025-10-03

Date: 2025-09-11

Generated from FC by FIHO.FI at 2025-10-08

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3. Directional Character
4. Feature Name
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6. Information
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9. Orientation
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1 Feature Types

1.1 Aids to Navigation

Type: Abstract FeatureType

Definition: A visual, acoustical, or radio device, external to a ship, designed to assist in determining a safe course or a vessel's position, or to warn of dangers and/or obstructions. Aids to navigation usually include buoys, beacons, fog signals, lights, radio beacons, leading marks, radio position fixing systems and GNSS which are chart-related and assist safe navigation.

CamelCase: AidsToNavigation

Super type: AbstractFeatureType

Primitive: -

Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
ID Code			TE	0, 1
Interoperability Identifier			UN	1, 1
Information			C	0, *
File Locator			(S) TE	0, 1
File Reference			(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1
Text			(S) TE	0, 1
Feature Name			C	0, *
Language			(S) TE	1, 1
Name			(S) TE	1, 1
Name Usage		1. Default Name Display 2. Alternate Name Display 3. No Chart Display	(S) EN	0, 1
Scale Minimum			IN	0, 1
Source Date			DA	0, 1
Source			TE	0, 1

Pictorial Representation			TE	0, 1
Installation Date			DA	0, 1
Fixed Date Range			C	0, 1
Date End			(S) TD	0, 1
Date Start			(S) TD	0, 1
Time of Day End			(S) TI	0, 1
Time of Day Start			(S) TI	0, 1
Periodic Date Range			C	0, 1
Date End			(S) TD	1, 1
Date Start			(S) TD	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	AtonAggregations	Aids to Navigation	atonAggregationBy	0..*	AtonAggregation	peerAtonAggregation	0..*
	--FeatureType-- Aids to Navigation 0..* <i>AtonAggregations</i> 0..* atonAggregationBy peerAtonAggregation --FeatureType-- AtonAggregation						
Association	AtonAssociations	Aids to Navigation	atonAssociationBy	0..*	AtonAssociation	peerAtonAssociation	0..*
	--FeatureType-- Aids to Navigation 0..* <i>AtonAssociations</i> 0..* atonAssociationBy peerAtonAssociation --FeatureType-- AtonAssociation						

Information bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	Atonstatus	Aids to Navigation			AtonStatusInformation	Statuspart	0..1
	--FeatureType-- Aids to Navigation <i>Atonstatus</i> 0..1 Statuspart --InformationType-- AtonStatusInformation						

1.2 Equipment

Type: FeatureType

Definition: The implements used in an operation or activity.

CamelCase: Equipment

Super type: AidsToNavigation

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	StructureEquipment	Equipment	child	0..*	StructureObject	parentFihoXyz	1..1
	--FeatureType-- Equipment		0..* child	StructureEquipment	1..1 parentFihoXyz	--FeatureType-- StructureObject	

1.3 Generic Buoy

Type: FeatureType

Definition: A floating object moored to the bottom in a particular (charted) place, as an aid to navigation or for other specific purposes.

CamelCase: GenericBuoy

Super type: StructureObject

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Buoy Shape		1. Conical 2. Can 3. Spherical 4. Pillar 5. Spar 6. Barrel 7. Superbuoy 8. Ice Buoy	EN	1, 1
Colour		1. White 2. Black 3. Red 4. Green	EN	1, *

		5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Radar Conspicuous			BO	0, 1
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. Main European Inland Waterway Marking System 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulation 15. Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction		1. Masonry	EN	0, *

		2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request	EN	0, *

		20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Type of Buoy			TE	0, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	BuoyTopmark	Generic Buoy	buoyPart	1..1	Topmark	topmarkPart	0..*
	--FeatureType-- Generic Buoy		1..1	<i>BuoyTopmark</i>	0..*	--FeatureType-- Topmark	
			buoyPart		topmarkPart		

1.4 Pile

Type: FeatureType

Definition: A long heavy timber or section of steel, wood, concrete, etc., forced into the earth or sea floor to serve as a support, as for a pier, or to resist lateral pressure; or as a free standing pole within a marine environment.

CamelCase: Pile

Super type: StructureObject

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Pile		1. Stake 3. Post 4. Tripodal 5. Piling 6. Area of Piles 7. Pipe 8. Mooring Post	EN	0, 1
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink	EN	0, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes	EN	0, *

		3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle		
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Height			RE	0, 1

1.5 Silo/Tank

Type: FeatureType

Definition: A large storage structure used for storing loose materials, liquids and/or gases.

CamelCase: SiloTank

Super type: StructureObject

Primitive: point,surface,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Building Shape		5. High-Rise Building 6. Pyramid 7. Cylindrical 8. Spherical 9. Cubic	EN	0, 1
Category of Silo/Tank		1. Silo in General 2. Tank in General 3. Grain Elevator 4. Water Tower	EN	0, 1
Colour		1. White	EN	0, *

		2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Radar Conspicuous			BO	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced	EN	0, *

		6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing	EN	0, *

		24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Height			RE	0, 1

1.6 Cardinal Buoy

Type: FeatureType

Definition: A cardinal buoy is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked.

CamelCase: CardinalBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Cardinal Mark		1. North Cardinal Mark 2. East Cardinal Mark	EN	1, 1

		3. South Cardinal Mark		
		4. West Cardinal Mark		

1.7 Emergency Wreck Marking Buoy

Type: FeatureType

Definition: An emergency wreck marking buoy is a buoy moored on or above a new wreck, designed to provide a prominent (both visual and radio) and easily identifiable temporary first response.

CamelCase: EmergencyWreckMarkingBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.8 Installation Buoy

Type: FeatureType

Definition: An installation buoy is a buoy used for loading tankers with gas or oil.

CamelCase: InstallationBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Installation Buoy		1. Catenary Anchor Leg Mooring 2. Single Buoy Mooring	EN	1, 1

1.9 Isolated Danger Buoy

Type: FeatureType

Definition: An isolated danger buoy is a buoy moored on or above an isolated danger of limited extent, which has navigable water all around it.

CamelCase: IsolatedDangerBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.10 Lateral Buoy

Type: FeatureType

Definition: A lateral buoy is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well-defined channels and are used in conjunction with a conventional direction of buoyage.

CamelCase: LateralBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Lateral Mark		1. Port-Hand Lateral Mark 2. Starboard-Hand Lateral Mark 3. Preferred Channel to Starboard Lateral Mark 4. Preferred Channel to Port Lateral Mark 5. Right-Hand Side of the Waterway 6. Left-Hand Side of the Waterway 7. Right-Hand Side of the Channel 8. Left-Hand Side of the Channel 9. Bifurcation of the Waterway 10. Bifurcation of the Channel 11. Channel Near the Right Bank 12. Channel Near the Left Bank 13. Channel Cross-Over to the Right Bank 14. Channel Cross-Over to the Left Bank 15. Danger Point or Obstacles at the Right-Hand Side 16. Danger Point or Obstacles at the Left-Hand Side 17. Turn Off at the Right-Hand Side 18. Turn Off at the Left-Hand Side 19. Junction at the Right-Hand Side	EN	1, 1

		20. Junction at the Left-Hand Side		
		21. Harbour Entry at the Right-Hand Side		
		22. Harbour Entry at the Left-Hand Side		
		23. Bridge Pier Mark		
		24. Entry From a Lake to a Narrower Waterway, Right Bank		
		25. Entry From a Lake to a Narrower Waterway, Left Bank		
		26. Change Bank		
		27. Continue Along Bank		

1.11 Light Float

Type: FeatureType

Definition: A boat-like structure used instead of a light buoy in waters where strong streams or currents are experienced, or when a greater elevation than that of a light buoy is necessary.

CamelCase: LightFloat

Super type: StructureObject

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta	EN	1, *

		13. Pink		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Radar Conspicuous			BO	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic	EN	0, *
Status		1. Permanent	EN	0, *

- | | | |
|--|---------------------------|--|
| | 2. Occasional | |
| | 3. Recommended | |
| | 4. Not in Use | |
| | 5. Periodic/Intermittent | |
| | 6. Reserved | |
| | 7. Temporary | |
| | 8. Private | |
| | 9. Mandatory | |
| | 11. Extinguished | |
| | 12. Illuminated | |
| | 13. Historic | |
| | 14. Public | |
| | 15. Synchronized | |
| | 16. Watched | |
| | 17. Unwatched | |
| | 18. Existence Doubtful | |
| | 19. On Request | |
| | 20. Drop Away | |
| | 21. Rising | |
| | 22. Increasing | |
| | 23. Decreasing | |
| | 24. Strong | |
| | 25. Good | |
| | 26. Moderately | |
| | 27. Poor | |
| | 28. Buoyed | |
| | 29. Fully Operational | |
| | 30. Partially Operational | |
| | 31. Drifting | |
| | 32. Broken | |
| | 33. Offline | |

		34. Discontinued		
		35. Manual Observation		
		36. Unknown Status		
		37. Confirmed		
		38. Candidate		
		39. Under Modification		
		41. Under Removal / Deletion		
		42. Removed / Deleted		
		43. Candidate for Modification		

1.12 Light Vessel

Type: FeatureType

Definition: A distinctively marked vessel anchored or moored at a charted point, to serve as an aid to navigation. By night, it displays a characteristic light(s) and is usually equipped with other devices, such as fog signal, submarine sound signal, and radio-beacon, to assist navigation.

CamelCase: LightVessel

Super type: StructureObject

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta	EN	1, *

		13. Pink		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Radar Conspicuous			BO	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic	EN	0, *
Status		1. Permanent	EN	0, *

- | | | |
|--|---------------------------|--|
| | 2. Occasional | |
| | 3. Recommended | |
| | 4. Not in Use | |
| | 5. Periodic/Intermittent | |
| | 6. Reserved | |
| | 7. Temporary | |
| | 8. Private | |
| | 9. Mandatory | |
| | 11. Extinguished | |
| | 12. Illuminated | |
| | 13. Historic | |
| | 14. Public | |
| | 15. Synchronized | |
| | 16. Watched | |
| | 17. Unwatched | |
| | 18. Existence Doubtful | |
| | 19. On Request | |
| | 20. Drop Away | |
| | 21. Rising | |
| | 22. Increasing | |
| | 23. Decreasing | |
| | 24. Strong | |
| | 25. Good | |
| | 26. Moderately | |
| | 27. Poor | |
| | 28. Buoyed | |
| | 29. Fully Operational | |
| | 30. Partially Operational | |
| | 31. Drifting | |
| | 32. Broken | |
| | 33. Offline | |

		34. Discontinued		
		35. Manual Observation		
		36. Unknown Status		
		37. Confirmed		
		38. Candidate		
		39. Under Modification		
		41. Under Removal / Deletion		
		42. Removed / Deleted		
		43. Candidate for Modification		

1.13 Mooring Buoy

Type: FeatureType

Definition: A buoy secured to the bottom by permanent moorings with means for mooring a vessel by use of its anchor chain or mooring lines.

CamelCase: MooringBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.14 Offshore Platform

Type: FeatureType

Definition: A permanent offshore structure, either fixed or floating.

CamelCase: OffshorePlatform

Super type: StructureObject

Primitive: point,surface,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Offshore Platform		1. Oil Rig 2. Production Platform 3. Observation/Research Platform 4. Articulated Loading Platform 5. Single Anchor Leg Mooring	EN	0, *

		6. Mooring Tower 7. Artificial Island 8. Floating Production, Storage and Off-Loading Vessel 9. Accommodation Platform 10. Navigation, Communication and Control Buoy 11. Floating Oil Tank		
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink	EN	0, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Condition		1. Under Construction	EN	0, 1

		2. Ruined 3. Under Reclamation 4. Wingless 5. Planned Construction		
Radar Conspicuous			BO	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic	EN	0, *
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private	EN	0, *

- | | | |
|--|------------------------------|--|
| | 9. Mandatory | |
| | 11. Extinguished | |
| | 12. Illuminated | |
| | 13. Historic | |
| | 14. Public | |
| | 15. Synchronized | |
| | 16. Watched | |
| | 17. Unwatched | |
| | 18. Existence Doubtful | |
| | 19. On Request | |
| | 20. Drop Away | |
| | 21. Rising | |
| | 22. Increasing | |
| | 23. Decreasing | |
| | 24. Strong | |
| | 25. Good | |
| | 26. Moderately | |
| | 27. Poor | |
| | 28. Buoyed | |
| | 29. Fully Operational | |
| | 30. Partially Operational | |
| | 31. Drifting | |
| | 32. Broken | |
| | 33. Offline | |
| | 34. Discontinued | |
| | 35. Manual Observation | |
| | 36. Unknown Status | |
| | 37. Confirmed | |
| | 38. Candidate | |
| | 39. Under Modification | |
| | 41. Under Removal / Deletion | |

		42. Removed / Deleted		
		43. Candidate for Modification		

1.15 Safe Water Buoy

Type: FeatureType

Definition: A safe water buoy is used to indicate that there is navigable water around the mark.

CamelCase: SafeWaterBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.16 Special Purpose/General Buoy

Type: FeatureType

Definition: A special purpose buoy is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners.

CamelCase: SpecialPurposeGeneralBuoy

Super type: GenericBuoy

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark		1. Firing Danger Area Mark 2. Target Mark 3. Marker Ship Mark 4. Degaussing Range Mark 5. Barge Mark 6. Cable Mark 7. Spoil Ground Mark 8. Outfall Mark 9. ODAS 10. Recording Mark 11. Seaplane Anchorage Mark 12. Recreation Zone Mark	EN	1, *

- | | | |
|--|--|--|
| | 13. Private Mark | |
| | 14. Mooring Mark | |
| | 15. LANBY | |
| | 16. Leading Mark | |
| | 17. Measured Distance Mark | |
| | 18. Notice Mark | |
| | 19. TSS Mark | |
| | 20. Anchoring Prohibited Mark | |
| | 21. Berthing Prohibited Mark | |
| | 22. Overtaking Prohibited Mark | |
| | 23. Two-Way Traffic Prohibited Mark | |
| | 24. Reduced Wake Mark | |
| | 25. Speed Limit Mark | |
| | 26. Stop Mark | |
| | 27. General Warning Mark | |
| | 28. Sound Ship's Siren Mark | |
| | 29. Restricted Vertical Clearance Mark | |
| | 30. Maximum Vessel's Draught Mark | |
| | 31. Restricted Horizontal Clearance Mark | |
| | 32. Strong Current Warning Mark | |
| | 33. Berthing Permitted Mark | |
| | 34. Overhead Power Cable Mark | |
| | 35. Channel Edge Gradient Mark | |
| | 36. Telephone Mark | |
| | 37. Ferry Crossing Mark | |
| | 39. Pipeline Mark | |
| | 40. Anchorage Mark | |
| | 41. Clearing Mark | |
| | 42. Control Mark | |
| | 43. Diving Mark | |

		44. Refuge Beacon		
		45. Foul Ground Mark		
		46. Yachting Mark		
		47. Heliport Mark		
		48. GNSS Mark		
		49. Seaplane Landing Mark		
		50. Entry Prohibited Mark		
		51. Work in Progress Mark		
		52. Mark With Unknown Purpose		
		53. Wellhead Mark		
		54. Channel Separation Mark		
		55. Marine Farm Mark		
		56. Artificial Reef Mark		
		57. Ice Mark		
		58. Nature Reserve Mark		
		59. Fish Aggregating Device		
		60. Wreck Mark		
		61. Customs Mark		
		62. Causeway Mark		
		63. Wave Recorder		
		64. Jetski Prohibited		

1.17 Navigation Line

Type: FeatureType

Definition: A straight line extending towards an area of navigational interest and generally generated by two navigational aids or one navigational aid and a bearing.

CamelCase: NavigationLine

Super type: AidsToNavigation

Primitive: curve,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Navigation Line		1. Clearing Line	EN	1, 1

		2. Transit Line 3. Leading Line Bearing a Recommended Track		
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately	EN	0, *

		27. Poor	
		28. Buoyed	
		29. Fully Operational	
		30. Partially Operational	
		31. Drifting	
		32. Broken	
		33. Offline	
		34. Discontinued	
		35. Manual Observation	
		36. Unknown Status	
		37. Confirmed	
		38. Candidate	
		39. Under Modification	
		41. Under Removal / Deletion	
		42. Removed / Deleted	
		43. Candidate for Modification	

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	RangeSystem	Navigation Line	navigationLineFihoXyz	1..*	RecommendedTrack	navigableTrack	0..*
	--FeatureType-- Navigation Line 1..* RangeSystem 0..* navigationLineFihoXyz navigableTrack --FeatureType-- RecommendedTrack						

1.18 Recommended Track

Type: FeatureType

Definition: A route which has been specially examined to ensure so far as possible that it is free of dangers and along which ships are advised to navigate.

CamelCase: RecommendedTrack

Super type: AidsToNavigation

Primitive: curve,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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Based On Fixed Marks			BO	1, 1
Depth Range Minimum Value			RE	0, 1
Maximal Permitted Draught			RE	0, 1
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1
Quality of Vertical Measurement		1. Depth Known 2. Depth or Least Depth Unknown 3. Doubtful Sounding 4. Unreliable Sounding 5. No Bottom Found at Value Shown 6. Least Depth Known 7. Least Depth Unknown, Safe Clearance at Value Shown 8. Value Reported (Not Surveyed) 9. Value Reported (Not Confirmed) 10. Maintained Depth 11. Not Regularly Maintained	EN	0, *
Vertical Uncertainty			C	0, 1
Uncertainty Fixed			(S) RE	1, 1
Uncertainty Variable Factor			(S) RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory	EN	0, *

- | | | |
|--|------------------------------|--|
| | 11. Extinguished | |
| | 12. Illuminated | |
| | 13. Historic | |
| | 14. Public | |
| | 15. Synchronized | |
| | 16. Watched | |
| | 17. Unwatched | |
| | 18. Existence Doubtful | |
| | 19. On Request | |
| | 20. Drop Away | |
| | 21. Rising | |
| | 22. Increasing | |
| | 23. Decreasing | |
| | 24. Strong | |
| | 25. Good | |
| | 26. Moderately | |
| | 27. Poor | |
| | 28. Buoyed | |
| | 29. Fully Operational | |
| | 30. Partially Operational | |
| | 31. Drifting | |
| | 32. Broken | |
| | 33. Offline | |
| | 34. Discontinued | |
| | 35. Manual Observation | |
| | 36. Unknown Status | |
| | 37. Confirmed | |
| | 38. Candidate | |
| | 39. Under Modification | |
| | 41. Under Removal / Deletion | |
| | 42. Removed / Deleted | |

		43. Candidate for Modification		
Technique of Vertical Measurement		1. Found by Echo Sounder 2. Found by Side Scan Sonar 3. Found by Multi Beam 4. Found by Diver 5. Found by Lead Line 6. Swept by Wire-Drag 7. Found by Laser 8. Swept by Vertical Acoustic System 9. Found by Electromagnetic Sensor 10. Photogrammetry 11. Satellite Imagery 12. Found by Levelling 13. Swept by Side Scan Sonar 14. Computer Generated 15. Found by LIDAR 16. Synthetic Aperture Radar 17. Hyperspectral Imagery 18. Mechanically Swept	EN	0, *
Traffic Flow		1. Inbound 2. Outbound 3. One-Way 4. Two-Way	EN	1, 1
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water	EN	0, 1

6. Lowest Low Water Springs

7. Approximate Mean Low Water
Springs

8. Indian Spring Low Water

9. Low Water Springs

10. Approximate Lowest
Astronomical Tide

11. Nearly Lowest Low Water

12. Mean Lower Low Water

13. Low Water

14. Approximate Mean Low Water

15. Approximate Mean Lower
Low Water

16. Mean High Water

17. Mean High Water Springs

18. High Water

19. Approximate Mean Sea Level

20. High Water Springs

21. Mean Higher High Water

22. Equinoctial Spring Low Water

23. Lowest Astronomical Tide

24. Local Datum

25. International Great Lakes
Datum 1985

26. Mean Water Level

27. Lower Low Water Large Tide

28. Higher High Water Large Tide

29. Nearly Highest High Water

30. Highest Astronomical Tide

31. Local Low Water Reference
Level

32. Local High Water Reference
Level

33. Local Mean Water Reference
Level

		34. Equivalent Height of Water (German GIW) 35. Highest Shipping Height of Water (German HSW) 36. Reference Low Water Level According to Danube Commission 37. Highest Shipping Height of Water According to Danube Commission 38. Dutch River Low Water Reference Level (OLR) 39. Russian Project Water Level 40. Russian Normal Backwater Level 41. Ohio River Datum 43. Dutch High Water Reference Level 44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW) 46. International Great Lakes Datum 2020 47. Sea Floor 48. Sea Surface 49. Hydrographic Zero		
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Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	RangeSystem	Recommended Track	navigableTrack	0..*	NavigationLine	navigationLineFihoXyz	1..*
	--FeatureType-- Recommended Track 0..* RangeSystem navigableTrack navigationLineFihoXyz 1..* --FeatureType-- NavigationLine						

1.19 Virtual AIS Aid to Navigation

Type: FeatureType

Definition: An Automatic Identification System (AIS) message 21 transmitted from an AIS station to simulate on navigation systems an Aid to Navigation which does not physically exist.

CamelCase: VirtualAISaidToNavigation

Super type: ElectronicAton

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Virtual AIS Aid to Navigation Type		1. North Cardinal 2. East Cardinal 3. South Cardinal 4. West Cardinal 5. Port Lateral 6. Starboard Lateral 7. Preferred Channel to Port 8. Preferred Channel to Starboard 9. Isolated Danger 10. Safe Water 11. Special Purpose 12. New Danger Marking	EN	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	VirtualAIS	Virtual AIS Aid to Navigation	virtualAISbroadcastBy	0..1	RadioStation	virtualAISbroadcasts	0..*
	--FeatureType-- Virtual AIS Aid to Navigation		0..1		VirtualAIS	0..*	--FeatureType-- RadioStation
					virtualAISbroadcastBy	virtualAISbroadcasts	

1.20 Daymark

Type: FeatureType

Definition: The identifying characteristics of an aid to navigation which serve to facilitate its recognition against a daylight viewing background. On those structures that do not by themselves present an adequate viewing area to be seen at the required distance, the aid is made more visible by affixing a daymark to the structure. A daymark so affixed has a distinctive colour and shape depending on the purpose of the aid.

CamelCase: Daymark

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark		1. Firing Danger Area Mark 2. Target Mark 3. Marker Ship Mark 4. Degaussing Range Mark 5. Barge Mark 6. Cable Mark 7. Spoil Ground Mark 8. Outfall Mark 9. ODAS 10. Recording Mark 11. Seaplane Anchorage Mark 12. Recreation Zone Mark 13. Private Mark 14. Mooring Mark 15. LANBY 16. Leading Mark 17. Measured Distance Mark 18. Notice Mark 19. TSS Mark 20. Anchoring Prohibited Mark 21. Berthing Prohibited Mark 22. Overtaking Prohibited Mark 23. Two-Way Traffic Prohibited Mark 24. Reduced Wake Mark 25. Speed Limit Mark 26. Stop Mark 27. General Warning Mark 28. Sound Ship's Siren Mark 29. Restricted Vertical Clearance Mark	EN	0, 1

30. Maximum Vessel's Draught Mark

31. Restricted Horizontal Clearance Mark

32. Strong Current Warning Mark

33. Berthing Permitted Mark

34. Overhead Power Cable Mark

35. Channel Edge Gradient Mark

36. Telephone Mark

37. Ferry Crossing Mark

39. Pipeline Mark

40. Anchorage Mark

41. Clearing Mark

42. Control Mark

43. Diving Mark

44. Refuge Beacon

45. Foul Ground Mark

46. Yachting Mark

47. Heliport Mark

48. GNSS Mark

49. Seaplane Landing Mark

50. Entry Prohibited Mark

51. Work in Progress Mark

52. Mark With Unknown Purpose

53. Wellhead Mark

54. Channel Separation Mark

55. Marine Farm Mark

56. Artificial Reef Mark

57. Ice Mark

58. Nature Reserve Mark

59. Fish Aggregating Device

60. Wreck Mark

61. Customs Mark

		62. Causeway Mark 63. Wave Recorder 64. Jetski Prohibited		
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink	EN	1, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Height			RE	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced	EN	0, *

		5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing	EN	0, *

		23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Topmark/Daymark Shape		1. Cone (Point Up) 2. Cone (Point Down) 3. Sphere 4. 2 Spheres 5. Cylinder 6. Board 7. X-Shaped 8. Upright Cross 9. Cube (Point Up) 10. 2 Cones (Point to Point) 11. 2 Cones (Base to Base)	EN	1, 1

		12. Rhombus 13. 2 Cones (Points Upward) 14. 2 Cones (Points Downward) 15. Besom (Point Up) 16. Besom (Point Down) 17. Flag 18. Sphere Over a Rhombus 19. Square 20. Rectangle (Horizontal) 21. Rectangle (Vertical) 22. Trapezium (Up) 23. Trapezium (Down) 24. Triangle (Point Up) 25. Triangle (Point Down) 26. Circle 27. Two Upright Crosses (One Over the Other) 28. T-Shape 29. Triangle Pointing Up Over a Circle 30. Upright Cross Over a Circle 31. Rhombus Over a Circle 32. Circle Over a Triangle Pointing Up 33. Other Shape (See Shape Information) 34. Tubular		
Orientation			C	0, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1

1.21 Structure Object

Type: FeatureType

Definition: Something (such as a house, tower, bridge, etc.) that is built by putting parts together and that usually stands on its own.

CamelCase: StructureObject

Super type: AidsToNavigation

Primitive: noGeometry,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
AtoN Number			TE	1, 1

Feature bindings:

Type	Association Name	Association Ends													
		Class	Role	Multiplicity	Class	Role	Multiplicity								
Association	StructureEquipment	Structure Object	parentFihoXyz	1..1	Equipment	child	0..*								
	<table><tr><td rowspan="2">--FeatureType-- Structure Object</td><td>1..1</td><td>StructureEquipment</td><td>0..*</td><td rowspan="2">--FeatureType-- Equipment</td></tr><tr><td>parentFihoXyz</td><td></td><td>child</td></tr></table>							--FeatureType-- Structure Object	1..1	StructureEquipment	0..*	--FeatureType-- Equipment	parentFihoXyz		child
	--FeatureType-- Structure Object	1..1	StructureEquipment	0..*	--FeatureType-- Equipment										
parentFihoXyz			child												

1.22 Fog Signal

Type: FeatureType

Definition: A warning signal transmitted by a vessel, or aid to navigation, during periods of low visibility. Also, the device producing such a signal.

CamelCase: FogSignal

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Fog Signal		1. Explosive 2. Diaphone 3. Siren 4. Nautophone 5. Reed 6. Tyfon 7. Bell 8. Whistle 9. Gong 10. Horn	EN	1, 1

Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken	EN	0, *
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		33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Signal Sequence			C	0, 1
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1

1.23 Radar Reflector

Type: FeatureType

Definition: A device capable of, or intended for, reflecting radar signals.

CamelCase: RadarReflector

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Height			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary	EN	0, *

- | |
|---------------------------|
| 8. Private |
| 9. Mandatory |
| 11. Extinguished |
| 12. Illuminated |
| 13. Historic |
| 14. Public |
| 15. Synchronized |
| 16. Watched |
| 17. Unwatched |
| 18. Existence Doubtful |
| 19. On Request |
| 20. Drop Away |
| 21. Rising |
| 22. Increasing |
| 23. Decreasing |
| 24. Strong |
| 25. Good |
| 26. Moderately |
| 27. Poor |
| 28. Buoyed |
| 29. Fully Operational |
| 30. Partially Operational |
| 31. Drifting |
| 32. Broken |
| 33. Offline |
| 34. Discontinued |
| 35. Manual Observation |
| 36. Unknown Status |
| 37. Confirmed |
| 38. Candidate |
| 39. Under Modification |

		41. Under Removal / Deletion		
		42. Removed / Deleted		
		43. Candidate for Modification		

1.24 Generic Beacon

Type: FeatureType

Definition: A fixed artificial navigation mark that can be recognized by its shape, colour, pattern, topmark or light character, or a combination of these. It may carry various additional aids to navigation.

CamelCase: GenericBeacon

Super type: StructureObject

Primitive: noGeometry,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Beacon Shape		1. Stake, Pole, Perch, Post 2. Withy 3. Beacon Tower 4. Lattice Beacon 5. Pile Beacon 6. Cairn 7. Buoyant Beacon	EN	1, 1
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange	EN	1, *

		12. Magenta 13. Pink		
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle	EN	0, *
Radar Conspicuous			BO	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	0, 1
Height			RE	0, 1
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. Main European Inland Waterway Marking System 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulation 15. Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders	EN	0, *

		4. Hard Surfaced 5. Unsurfaced 6. Wooden 7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Vertical Length			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away	EN	0, *

		21. Rising		
		22. Increasing		
		23. Decreasing		
		24. Strong		
		25. Good		
		26. Moderately		
		27. Poor		
		28. Buoyed		
		29. Fully Operational		
		30. Partially Operational		
		31. Drifting		
		32. Broken		
		33. Offline		
		34. Discontinued		
		35. Manual Observation		
		36. Unknown Status		
		37. Confirmed		
		38. Candidate		
		39. Under Modification		
		41. Under Removal / Deletion		
		42. Removed / Deleted		
		43. Candidate for Modification		

1.25 Radar Transponder Beacon

Type: FeatureType

Definition: A transponder beacon transmitting a coded signal on radar frequency, permitting an interrogating craft to determine the bearing and range of the transponder.

CamelCase: RadarTransponderBeacon

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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Category of Radar Transponder Beacon		1. Ramark, Radar Beacon Transmitting Continuously 2. Racon, Radar Transponder Beacon 3. Leading Racon/Radar Transponder Beacon	EN	1, 1
Radar Wave Length			C	0, 1
Radar Band			(S) TE	1, 1
Wave Length Value			(S) RE	1, 1
Sector Limit One			C	0, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1
Sector Limit Two			C	0, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1
Signal Group			TE	0, 1
Signal Sequence			C	0, 1
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated	EN	0, *

		13. Historic		
		14. Public		
		15. Synchronized		
		16. Watched		
		17. Unwatched		
		18. Existence Doubtful		
		19. On Request		
		20. Drop Away		
		21. Rising		
		22. Increasing		
		23. Decreasing		
		24. Strong		
		25. Good		
		26. Moderately		
		27. Poor		
		28. Buoyed		
		29. Fully Operational		
		30. Partially Operational		
		31. Drifting		
		32. Broken		
		33. Offline		
		34. Discontinued		
		35. Manual Observation		
		36. Unknown Status		
		37. Confirmed		
		38. Candidate		
		39. Under Modification		
		41. Under Removal / Deletion		
		42. Removed / Deleted		
		43. Candidate for Modification		
Value of Nominal Range			RE	0, 1

1.26 Radio Station

Type: FeatureType

Definition: A place equipped to transmit radio waves. Such a station may be either stationary or mobile, and may also be provided with a radio receiver.

CamelCase: RadioStation

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Radio Station		1. Circular (Non-Directional) Marine or Aero-Marine Radiobeacon 2. Directional Radiobeacon 3. Rotating Pattern Radiobeacon 4. Consol Beacon 5. Radio Direction-Finding Station 6. Coast Radio Station Providing QTG Service 7. Aeronautical Radiobeacon 8. Decca 9. Loran C 10. Differential GNSS 11. Toran 12. Omega 13. Syledis 14. Chaika 19. Radio Telephone Station 20. AIS Base Station	EN	1, 1
Estimated Range of Transmission			RE	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use	EN	0, *

5. Periodic/Intermittent

6. Reserved

7. Temporary

8. Private

9. Mandatory

11. Extinguished

12. Illuminated

13. Historic

14. Public

15. Synchronized

16. Watched

17. Unwatched

18. Existence Doubtful

19. On Request

20. Drop Away

21. Rising

22. Increasing

23. Decreasing

24. Strong

25. Good

26. Moderately

27. Poor

28. Buoyed

29. Fully Operational

30. Partially Operational

31. Drifting

32. Broken

33. Offline

34. Discontinued

35. Manual Observation

36. Unknown Status

		37. Confirmed	
		38. Candidate	
		39. Under Modification	
		41. Under Removal / Deletion	
		42. Removed / Deleted	
		43. Candidate for Modification	

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	PhysicalAIS	Radio Station	physicalAIS broadcasts	0..*	PhysicalAIS AidToNavigation	physicalAIS broadcastBy	0..1
	--FeatureType-- Radio Station 0..* PhysicalAIS 0..1 physicalAISbroadcasts physicalAISbroadcastBy --FeatureType-- PhysicalAISAidToNavigation						
Association	SyntheticAIS	Radio Station	syntheticAIS broadcasts	0..*	SyntheticAIS AidToNavigation	syntheticAIS broadcastBy	0..1
	--FeatureType-- Radio Station 0..* SyntheticAIS 0..1 syntheticAISbroadcasts syntheticAISbroadcastBy --FeatureType-- SyntheticAISAidToNavigation						
Association	VirtualAIS	Radio Station	virtualAIS broadcasts	0..*	VirtualAIS AidToNavigation	virtualAIS broadcastBy	0..1
	--FeatureType-- Radio Station 0..* VirtualAIS 0..1 virtualAISbroadcasts virtualAISbroadcastBy --FeatureType-- VirtualAISAidToNavigation						

1.27 Light Air Obstruction

Type: FeatureType

Definition: An air obstruction light is a light marking an obstacle which constitutes a danger to air navigation.

CamelCase: LightAirObstruction

Super type: GenericLight

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White	EN	1, *

		2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink		
Light Visibility		1. High Intensity 2. Low Intensity 3. Faint 4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range	EN	0, *
Exhibition Condition of Light		1. Light Shown Without Change of Character 2. Daytime Light 3. Fog Light 4. Night Light	EN	0, 1
Value of Nominal Range			RE	0, 1
Flare Bearing			IN	0, 1
Multiplicity of Features			C	0, 1
Multiplicity Known			(S) BO	1, 1

Number of Features			(S) IN	0, 1
Rhythm of Light			C	1, 1
Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 9. Interrupted Quick Flashing 10. Interrupted Very Quick Flashing 11. Interrupted Ultra Quick-Flashing 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing	(S) EN	1, 1
Signal Group			(S) TE	0, 10

Signal Period			(S) RE	0, 1
Signal Sequence			(S) C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1

1.28 Retroreflector

Type: FeatureType

Definition: A means of distinguishing unlighted marks at night. Retro-reflective material is secured to the mark in a particular pattern to reflect back light.

CamelCase: Retroreflector

Super type: Equipment

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink	EN	0, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes	EN	0, *

		4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle 9. Triangle		
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. Main European Inland Waterway Marking System 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulation 15. Paraguay-Parana Waterway - Brazilian Complementary Aids	EN	0, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched	EN	0, *

		17. Unwatched	
		18. Existence Doubtful	
		19. On Request	
		20. Drop Away	
		21. Rising	
		22. Increasing	
		23. Decreasing	
		24. Strong	
		25. Good	
		26. Moderately	
		27. Poor	
		28. Buoyed	
		29. Fully Operational	
		30. Partially Operational	
		31. Drifting	
		32. Broken	
		33. Offline	
		34. Discontinued	
		35. Manual Observation	
		36. Unknown Status	
		37. Confirmed	
		38. Candidate	
		39. Under Modification	
		41. Under Removal / Deletion	
		42. Removed / Deleted	
		43. Candidate for Modification	

1.29 Light All Around

Type: FeatureType

Definition: An all around light is a light that is visible over the whole horizon of interest to marine navigation and having no change in the characteristics of the light.

CamelCase: LightAllAround

Super type: GenericLight

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink	EN	1, *
Signal Generation		1. Automatically 2. By Wave Action 3. By Hand 4. By Wind 5. Radio Activated 6. Call Activated	EN	0, 1
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System 11. Main European Inland Waterway Marking System 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulation	EN	0, 1

		15. Paraguay-Parana Waterway - Brazilian Complementary Aids		
Major Light			BO	0, 1
Light Visibility		1. High Intensity 2. Low Intensity 3. Faint 4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range	EN	0, 1
Exhibition Condition of Light		1. Light Shown Without Change of Character 2. Daytime Light 3. Fog Light 4. Night Light	EN	0, 1
Category of Light		1. Directional Function 4. Leading Light 5. Aero Light 6. Air Obstruction Light 8. Flood Light 9. Strip Light 10. Subsidiary Light 11. Spotlight 12. Front 13. Rear 14. Lower 15. Upper 17. Emergency 18. Bearing Light	EN	0, *

		19. Horizontally Disposed 20. Vertically Disposed		
Value of Nominal Range			RE	0, 1
Multiplicity of Features			C	0, 1
Multiplicity Known			(S) BO	1, 1
Number of Features			(S) IN	0, 1
Rhythm of Light			C	1, 1
Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 9. Interrupted Quick Flashing 10. Interrupted Very Quick Flashing 11. Interrupted Ultra Quick-Flashing 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash	(S) EN	1, 1

		27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing		
Signal Group			(S) TE	0, 10
Signal Period			(S) RE	0, 1
Signal Sequence			(S) C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1

1.30 Light Fog Detector

Type: FeatureType

Definition: A fog detector light is a light used to automatically determine conditions of visibility which warrant the turning on or off of a sound signal.

CamelCase: LightFogDetector

Super type: GenericLight

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Rhythm of Light			C	1, 1
Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 9. Interrupted Quick Flashing 10. Interrupted Very Quick Flashing	(S) EN	1, 1

		11. Interrupted Ultra Quick-Flashing 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing		
Signal Group			(S) TE	0, 10
Signal Period			(S) RE	0, 1
Signal Sequence			(S) C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1
Signal Generation		1. Automatically 2. By Wave Action 3. By Hand 4. By Wind 5. Radio Activated 6. Call Activated	EN	0, 1

1.31 Light Sected

Type: FeatureType

Definition: A light presenting different appearances (in particular, different colours) over various parts of the horizon of interest to maritime navigation.

CamelCase: LightSectored

Super type: GenericLight

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink	EN	1, *
Signal Generation		1. Automatically 2. By Wave Action 3. By Hand 4. By Wind 5. Radio Activated 6. Call Activated	EN	0, 1
Marks Navigational - System Of		1. IALA A 2. IALA B 9. No System 10. Other System	EN	0, 1

		11. Main European Inland Waterway Marking System 12. Russian Inland Waterway Regulations 13. Brazilian National Inland Waterway Regulation 15. Paraguay-Parana Waterway - Brazilian Complementary Aids		
Exhibition Condition of Light		1. Light Shown Without Change of Character 2. Daytime Light 3. Fog Light 4. Night Light	EN	0, 1
Category of Light		1. Directional Function 4. Leading Light 5. Aero Light 6. Air Obstruction Light 8. Flood Light 9. Strip Light 10. Subsidiary Light 11. Spotlight 12. Front 13. Rear 14. Lower 15. Upper 17. Emergency 18. Bearing Light 19. Horizontally Disposed 20. Vertically Disposed	EN	0, *
Sector Characteristics			C	1, *
Light Characteristic		1. Fixed 2. Flashing	(S) EN	1, 1

		3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 9. Interrupted Quick Flashing 10. Interrupted Very Quick Flashing 11. Interrupted Ultra Quick-Flashing 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing		
Light Sector			(S) C	1, 10
Colour		1. White 2. Black 3. Red 4. Green	(S) EN	1, 99

		5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink		
Directional Character			(S) C	0, 1
Moire Effect			(S) BO	0, 1
Orientation			(S) C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1
Light Visibility		1. High Intensity 2. Low Intensity 3. Faint 4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range	(S) EN	0, 99
Sector Limit			(S) C	0, 1
Sector Limit One			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1
Sector Limit Two			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1

Value of Nominal Range			(S) RE	0, 1
Sector Information			(S) C	0, 99
Language			(S) TE	0, 1
Text			(S) TE	1, 1
Sector Arc Extension			(S) BO	0, 1
Signal Group			(S) TE	0, 10
Signal Period			(S) RE	0, 1
Signal Sequence			(S) C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1
Multiplicity of Features			C	0, 1
Multiplicity Known			(S) BO	1, 1
Number of Features			(S) IN	0, 1

1.32 Cardinal Beacon

Type: FeatureType

Definition: A cardinal beacon is used in conjunction with the compass to indicate where the mariner may find the best navigable water. It is placed in one of the four quadrants (North, East, South and West), bounded by inter-cardinal bearings from the point marked.

CamelCase: CardinalBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Cardinal Mark		1. North Cardinal Mark 2. East Cardinal Mark 3. South Cardinal Mark 4. West Cardinal Mark	EN	1, 1

1.33 Isolated Danger Beacon

Type: FeatureType

Definition: An isolated danger beacon is a beacon erected on an isolated danger of limited extent, which has navigable water all around it.

CamelCase: IsolatedDangerBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.34 Landmark

Type: FeatureType

Definition: A prominent object at a fixed location on land which can be used in determining a location or a direction.

CamelCase: Landmark

Super type: StructureObject

Primitive: point,curve,surface,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Landmark		1. Cairn 2. Cemetery 3. Chimney 4. Dish Aerial 5. Flagstaff 6. Flare Stack 7. Mast 8. Windsock 9. Monument 10. Column/Pillar 11. Memorial Plaque 12. Obelisk 13. Statue 14. Cross 15. Dome 16. Radar Scanner 17. Tower 18. Windmill	EN	1, *

		19. Windmotor 20. Spire/Minaret 21. Large Rock or Boulder on Land 22. Triangulation Mark 23. Boundary Mark 24. Observation Wheel 25. Torii 26. Bridge 27. Dam		
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink	EN	0, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle	EN	0, *

		9. Triangle		
Radar Conspicuous			BO	0, 1
Visual Prominence		1. Visually Conspicuous 2. Not Visually Conspicuous 3. Prominent	EN	1, 1
Function		2. Harbour-Masters Office 3. Customs Office 4. Health Office 5. Hospital 6. Post Office 7. Hotel 8. Railway Station 9. Police Station 10. Water-Police Station 11. Pilot Office 12. Pilot Lookout 13. Bank Office 14. Headquarters for District Control 15. Transit Shed/Warehouse 16. Factory 17. Power Station 18. Administrative 19. Educational Facility 20. Church 21. Chapel 22. Temple 23. Pagoda 24. Shinto Shrine 25. Buddhist Temple 26. Mosque	EN	0, *

		27. Marabout 28. Lookout 29. Communication 30. Television 31. Radio 32. Radar 33. Light Support 34. Microwave 35. Cooling 36. Observation 37. Timeball 38. Clock 39. Control 40. Airship Mooring 41. Stadium 42. Bus Station 43. Passenger Terminal Building 44. Sea Rescue Control 45. Observatory 46. Ore Crusher 47. Boathouse 48. Pumping Station 49. Roof Above Navigable Water 50. Building Above Navigable Water		
Nature of Construction		1. Masonry 2. Concreted 3. Loose Boulders 4. Hard Surfaced 5. Unsurfaced 6. Wooden	EN	0, *

		7. Metal 8. Glass Reinforced Plastic 9. Painted 10. Framework 11. Latticed 12. Glass 13. Fiberglass 14. Plastic		
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong	EN	0, *

		25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Height			RE	0, 1

1.35 Lateral Beacon

Type: FeatureType

Definition: A lateral beacon is used to indicate the port or starboard hand side of the route to be followed. They are generally used for well defined channels and are used in conjunction with a conventional direction of buoyage.

CamelCase: LateralBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Lateral Mark		1. Port-Hand Lateral Mark 2. Starboard-Hand Lateral Mark 3. Preferred Channel to Starboard Lateral Mark	EN	1, 1

		4. Preferred Channel to Port Lateral Mark	
		5. Right-Hand Side of the Waterway	
		6. Left-Hand Side of the Waterway	
		7. Right-Hand Side of the Channel	
		8. Left-Hand Side of the Channel	
		9. Bifurcation of the Waterway	
		10. Bifurcation of the Channel	
		11. Channel Near the Right Bank	
		12. Channel Near the Left Bank	
		13. Channel Cross-Over to the Right Bank	
		14. Channel Cross-Over to the Left Bank	
		15. Danger Point or Obstacles at the Right-Hand Side	
		16. Danger Point or Obstacles at the Left-Hand Side	
		17. Turn Off at the Right-Hand Side	
		18. Turn Off at the Left-Hand Side	
		19. Junction at the Right-Hand Side	
		20. Junction at the Left-Hand Side	
		21. Harbour Entry at the Right-Hand Side	
		22. Harbour Entry at the Left-Hand Side	
		23. Bridge Pier Mark	
		24. Entry From a Lake to a Narrower Waterway, Right Bank	
		25. Entry From a Lake to a Narrower Waterway, Left Bank	
		26. Change Bank	
		27. Continue Along Bank	

Type: FeatureType

Definition: A distinctive structure on or off a coast exhibiting a major light designed to serve as an aid to navigation.

CamelCase: Lighthouse

Super type: Landmark

Primitive: point,surface,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.37 Safe Water Beacon

Type: FeatureType

Definition: A safe water beacon is used to indicate that there is navigable water around the mark.

CamelCase: SafeWaterBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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1.38 Special Purpose/General Beacon

Type: FeatureType

Definition: A special purpose beacon is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notices to Mariners.

CamelCase: SpecialPurposeGeneralBeacon

Super type: GenericBeacon

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Special Purpose Mark		1. Firing Danger Area Mark 2. Target Mark 3. Marker Ship Mark 4. Degaussing Range Mark 5. Barge Mark 6. Cable Mark 7. Spoil Ground Mark	EN	1, *

- | | | |
|--|--|--|
| | 8. Outfall Mark | |
| | 9. ODAS | |
| | 10. Recording Mark | |
| | 11. Seaplane Anchorage Mark | |
| | 12. Recreation Zone Mark | |
| | 13. Private Mark | |
| | 14. Mooring Mark | |
| | 15. LANBY | |
| | 16. Leading Mark | |
| | 17. Measured Distance Mark | |
| | 18. Notice Mark | |
| | 19. TSS Mark | |
| | 20. Anchoring Prohibited Mark | |
| | 21. Berthing Prohibited Mark | |
| | 22. Overtaking Prohibited Mark | |
| | 23. Two-Way Traffic Prohibited Mark | |
| | 24. Reduced Wake Mark | |
| | 25. Speed Limit Mark | |
| | 26. Stop Mark | |
| | 27. General Warning Mark | |
| | 28. Sound Ship's Siren Mark | |
| | 29. Restricted Vertical Clearance Mark | |
| | 30. Maximum Vessel's Draught Mark | |
| | 31. Restricted Horizontal Clearance Mark | |
| | 32. Strong Current Warning Mark | |
| | 33. Berthing Permitted Mark | |
| | 34. Overhead Power Cable Mark | |
| | 35. Channel Edge Gradient Mark | |
| | 36. Telephone Mark | |
| | 37. Ferry Crossing Mark | |

		39. Pipeline Mark	
		40. Anchorage Mark	
		41. Clearing Mark	
		42. Control Mark	
		43. Diving Mark	
		44. Refuge Beacon	
		45. Foul Ground Mark	
		46. Yachting Mark	
		47. Heliport Mark	
		48. GNSS Mark	
		49. Seaplane Landing Mark	
		50. Entry Prohibited Mark	
		51. Work in Progress Mark	
		52. Mark With Unknown Purpose	
		53. Wellhead Mark	
		54. Channel Separation Mark	
		55. Marine Farm Mark	
		56. Artificial Reef Mark	
		57. Ice Mark	
		58. Nature Reserve Mark	
		59. Fish Aggregating Device	
		60. Wreck Mark	
		61. Customs Mark	
		62. Causeway Mark	
		63. Wave Recorder	
		64. Jetski Prohibited	

1.39 Dangerous Feature

Type: FeatureType

Definition: -

CamelCase: DangerousFeature

Super type: AbstractFeatureType

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Information			C	0, *
File Locator			(S) TE	0, 1
File Reference			(S) TE	0, 1
Headline			(S) TE	0, 1
Language			(S) TE	1, 1
Text			(S) TE	0, 1
Interoperability Identifier			UN	0, 1

Feature bindings:

Type	Association Name	Association Ends														
		Class	Role	Multiplicity	Class	Role	Multiplicity									
Association	DangerousFeatureAssociation	Dangerous Feature	danger	0..*	AtonAssociation	markingAton	1..*									
	<table><tr><td>--FeatureType-- Dangerous Feature</td><td>0..</td><td><i>DangerousFeatureAssociation</i></td><td>1..</td><td>--FeatureType-- AtonAssociation</td></tr><tr><td></td><td></td><td>danger</td><td></td><td>markingAton</td></tr></table>							--FeatureType-- Dangerous Feature	0..	<i>DangerousFeatureAssociation</i>	1..	--FeatureType-- AtonAssociation			danger	
--FeatureType-- Dangerous Feature	0..	<i>DangerousFeatureAssociation</i>	1..	--FeatureType-- AtonAssociation												
		danger		markingAton												

1.40 Aton Association

Type: FeatureType

Definition: Used to identify an association between two or more objects. The association may be named content of categoryOfAssociation should be put in information attribute when converting to S-57

CamelCase: AtonAssociation

Super type: AbstractFeatureType

Primitive: noGeometry,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Association		1. Channel Markings 2. Danger Markings	CO	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity

Association	DangerousFeatureAssociation	Aton Association	markingAton	1..*	DangerousFeature	danger	0..*
	--FeatureType-- Aton Association			1..* DangerousFeatureAssociation markingAton	--FeatureType-- DangerousFeature		
Association	AtonAssociations	Aton Association	peerAtonAssociation	0..*	AidsToNavigation	atonAssociationBy	0..*
	--FeatureType-- Aton Association			0..* AtonAssociations peerAtonAssociation atonAssociationBy	--FeatureType-- AidsToNavigation		

1.41 Aton Aggregation

Type: FeatureType

Definition: Used to identify an aggregation of two or more objects. This aggregation may be named content of categoryOfAggregation should be put in information attribute when converting to S-57.

CamelCase: AtonAggregation

Super type: AbstractFeatureType

Primitive: noGeometry,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category of Aggregation		1. leading line 3. measured distance 2. range system	CO	1, 1

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	AtonAggregations	Aton Aggregation	peerAtonAggregation	0..*	AidsToNavigation	atonAggregationBy	0..*
	--FeatureType-- Aton Aggregation			0..* AtonAggregations peerAtonAggregation atonAggregationBy	--FeatureType-- AidsToNavigation		

1.42 Electronic Aton

Type: Abstract FeatureType

Definition: TBD

CamelCase: ElectronicAton

Super type: AidsToNavigation

Primitive: -

Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
AtoN Number			TE	0, 1
MMSI Code			TE	1, 1
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor	EN	0, *

		28. Buoyed		
		29. Fully Operational		
		30. Partially Operational		
		31. Drifting		
		32. Broken		
		33. Offline		
		34. Discontinued		
		35. Manual Observation		
		36. Unknown Status		
		37. Confirmed		
		38. Candidate		
		39. Under Modification		
		41. Under Removal / Deletion		
		42. Removed / Deleted		
		43. Candidate for Modification		

1.43 Generic Light

Type: Abstract FeatureType

Definition: -

CamelCase: GenericLight

Super type: Equipment

Primitive: -

Feature use type: -

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Height			RE	0, 1
Vertical Length			RE	0, 1
Vertical Datum		1. Mean Low Water Springs 2. Mean Lower Low Water Springs 3. Mean Sea Level 4. Lowest Low Water 5. Mean Low Water	EN	0, 1

6. Lowest Low Water Springs

7. Approximate Mean Low Water Springs

8. Indian Spring Low Water

9. Low Water Springs

10. Approximate Lowest Astronomical Tide

11. Nearly Lowest Low Water

12. Mean Lower Low Water

13. Low Water

14. Approximate Mean Low Water

15. Approximate Mean Lower Low Water

16. Mean High Water

17. Mean High Water Springs

18. High Water

19. Approximate Mean Sea Level

20. High Water Springs

21. Mean Higher High Water

22. Equinoctial Spring Low Water

23. Lowest Astronomical Tide

24. Local Datum

25. International Great Lakes Datum 1985

26. Mean Water Level

27. Lower Low Water Large Tide

28. Higher High Water Large Tide

29. Nearly Highest High Water

30. Highest Astronomical Tide

31. Local Low Water Reference Level

32. Local High Water Reference Level

33. Local Mean Water Reference Level

		34. Equivalent Height of Water (German GIW) 35. Highest Shipping Height of Water (German HSW) 36. Reference Low Water Level According to Danube Commission 37. Highest Shipping Height of Water According to Danube Commission 38. Dutch River Low Water Reference Level (OLR) 39. Russian Project Water Level 40. Russian Normal Backwater Level 41. Ohio River Datum 43. Dutch High Water Reference Level 44. Baltic Sea Chart Datum 2000 45. Dutch Estuary Low Water Reference Level (OLW) 46. International Great Lakes Datum 2020 47. Sea Floor 48. Sea Surface 49. Hydrographic Zero		
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated	EN	0, *

		13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting 32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Effective Intensity			RE	0, 1

Peak Intensity			RE	0, 1
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1.44 Topmark

Type: FeatureType

Definition: A characteristic shape secured at the top of a buoy or beacon to aid in its identification.

CamelCase: Topmark

Super type: AidsToNavigation

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink	EN	0, *
Colour Pattern		1. Horizontal Stripes 2. Vertical Stripes 3. Diagonal Stripes 4. Squared 5. Stripes (Direction Unknown) 6. Border Stripe 7. Single Colour 8. Rectangle	EN	0, *

		9. Triangle		
Status		1. Permanent 2. Occasional 3. Recommended 4. Not in Use 5. Periodic/Intermittent 6. Reserved 7. Temporary 8. Private 9. Mandatory 11. Extinguished 12. Illuminated 13. Historic 14. Public 15. Synchronized 16. Watched 17. Unwatched 18. Existence Doubtful 19. On Request 20. Drop Away 21. Rising 22. Increasing 23. Decreasing 24. Strong 25. Good 26. Moderately 27. Poor 28. Buoyed 29. Fully Operational 30. Partially Operational 31. Drifting	EN	0, *

		32. Broken 33. Offline 34. Discontinued 35. Manual Observation 36. Unknown Status 37. Confirmed 38. Candidate 39. Under Modification 41. Under Removal / Deletion 42. Removed / Deleted 43. Candidate for Modification		
Topmark/Daymark Shape		1. Cone (Point Up) 2. Cone (Point Down) 3. Sphere 4. 2 Spheres 5. Cylinder 6. Board 7. X-Shaped 8. Upright Cross 9. Cube (Point Up) 10. 2 Cones (Point to Point) 11. 2 Cones (Base to Base) 12. Rhombus 13. 2 Cones (Points Upward) 14. 2 Cones (Points Downward) 15. Besom (Point Up) 16. Besom (Point Down) 17. Flag 18. Sphere Over a Rhombus 19. Square 20. Rectangle (Horizontal)	EN	1, 1

		21. Rectangle (Vertical) 22. Trapezium (Up) 23. Trapezium (Down) 24. Triangle (Point Up) 25. Triangle (Point Down) 26. Circle 27. Two Upright Crosses (One Over the Other) 28. T-Shape 29. Triangle Pointing Up Over a Circle 30. Upright Cross Over a Circle 31. Rhombus Over a Circle 32. Circle Over a Triangle Pointing Up 33. Other Shape (See Shape Information) 34. Tubular	
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Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	BuoyTopmark	Topmark	topmarkPart	0..*	GenericBuoy	buoyPart	1..1
	--FeatureType-- Topmark 0..* topmarkPart BuoyTopmark 1..1 buoyPart --FeatureType-- GenericBuoy						

1.45 Physical AIS Aid to Navigation

Type: FeatureType

Definition: An Automatic Identification System (AIS) message 21 transmitted from a physical Aid to Navigation, or transmitted from an AIS station for an Aid to Navigation which physically exists.

CamelCase: PhysicalAIS AidToNavigation

Super type: ElectronicAton

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
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Category Of Physical AIS Aid To Navigation		1. Physical AIS Type 1 2. Physical AIS Type 2 3. Physical AIS Type 3	EN	1, 1
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Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	PhysicalAIS	Physical AIS Aid to Navigation	physicalAIS broadcastBy	0..1	RadioStation	physicalAIS broadcasts	0..*
	--FeatureType-- Physical AIS Aid to Navigation 0..1PhysicalAISbroadcastByphysicalAISbroadcasts0..*--FeatureType-- RadioStation						

1.46 Synthetic AIS Aid to Navigation

Type: FeatureType

Definition: An Automatic Identification System (AIS) message 21 transmitted from an AIS station located remotely from the intended physical Aid to Navigation.

CamelCase: SyntheticAIS AidToNavigation

Super type: ElectronicAton

Primitive: point,

Feature use type: geographic

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Category Of Synthetic AIS Aid To Navigation		1. predicted 2. monitored	EN	1, 1
Virtual AIS Aid to Navigation Type		1. North Cardinal 2. East Cardinal 3. South Cardinal 4. West Cardinal 5. Port Lateral 6. Starboard Lateral 7. Preferred Channel to Port 8. Preferred Channel to Starboard 9. Isolated Danger	EN	1, 1

		10. Safe Water		
		11. Special Purpose		
		12. New Danger Marking		

Feature bindings:

Type	Association Name	Association Ends					
		Class	Role	Multiplicity	Class	Role	Multiplicity
Association	SyntheticAIS	Synthetic AIS Aid to Navigation	syntheticAIS broadcastBy	0..1	RadioStation	syntheticAIS broadcasts	0..*
	--FeatureType-- Synthetic AIS Aid to Navigation		0..1	SyntheticAIS	0..*	--FeatureType-- RadioStation	

4 Information Types

4.1 Aton Status Information

Type: InformationType

Definition: -

CamelCase: AtonStatusInformation

Attribute bindings:

S-10x Attribute	S-57 Acronym	Allowable Encoding Value	Type	Multiplicity
Change Details			C	1, 1
Aton Commissioning		1. Buoy Establishment 2. Light Establishment 3. Beacon Establishment 4. Audible Signal Establishment 5. Fog Signal Establishment 6. AIS Transmitter Establishment 7. V-AIS Establishment 8. RACON Establishment 9. DGPS Station Establishment 10. ELORAN Station Establishment 11. DGLONASS Station Establishment	(S) EN	0, 1

		12. E-Chayka Station Establishment		
		13. EGNOS establishment		
Aton Removal		1. Buoy Removal 2. Buoy Temporary Removal 3. Light Removal 4. Light Temporary Removal 5. Beacon Removal 6. Beacon Temporary Removal 7. Fog Signal Removal 8. Fog Signal Temporary Removal 9. Audible Signal Removal 10. Audible Signal Temporary Removal 11. V-AIS Removal 12. V-AIS Temporary Removal 13. RACON signal removal 14. RACON Temporary Removal 15. DGPS removal 16. DGPS temporary removal 17. EGNOS removal 18. EGNOS temporary removal 19. LORAN C Station Removal 20. LORAN C Station Temporary Removal 21. eLORAN removal 22. eLORAN temporary removal 23. Chayka Station Removal 24. Chayka Station Temporary Removal 25. E-Chayka Station Removal 26. E-Chayka Station Temporary Removal	(S) EN	0, 1

Aton Replacement		1. Buoy Change 2. Buoy Temporary Change 3. Light Change 4. Light Temporary Change 5. Sector Light Change 6. Sector Light Temporary Change 7. Beacon Change 8. Beacon Temporary Change 9. Fog Signal Change 10. Fog Signal Temporary Change 11. Audible Signal Change 12. Audible Signal Temporary Change 13. V-AIS Change 14. V-AIS Temporary Change 15. RACON signal change 16. RACON Temporary Change	(S) EN	0, 1
Fixed Aton Change		1. Beacon Missing 2. Beacon Damaged 3. Light beacon Unlit 4. Light beacon Unreliable 5. Light beacon Not synchronized 6. Light beacon damaged 7. Beacon Topmark Missing 8. Beacon Topmark Damaged 9. Beacon Daymark Unreliable 10. Floodlit beacon Unlit 11. Beacon Restored To Normal	(S) EN	0, 1
Floating Aton Change		1. Buoy Adrift 2. Buoy Damaged	(S) EN	0, 1

		3. Buoy Daymark Unreliable 4. Buoy Destroyed 5. Buoy Missing 6. Buoy Move 7. Buoy off Position 8. Buoy re-establishment 9. Buoy Restored to Normal 10. Buoy Topmark Damaged 11. Buoy Topmark Missing 12. Buoy Will Be Withdrawn 13. Buoy Withdrawn 14. Decommissioned for winter 15. Lifted for Winter 16. Light buoy Light damaged 17. Light buoy Light not synchronized 18. Light buoy Light unlit 19. Light buoy Light unreliable 20. Marine Aids to Navigation Unreliable 21. Recommissioned for navigation season 22. Replaced by Winter Spar 23. Seasonal Decommissioning Complete 24. Seasonal Decommissioning in Progress 25. Seasonal recommissioning complete 26. Seasonal recommissioning in progress		
Audible Signal Aton Change		1. Audible Signal Out Of Service 2. Fog Signal Out Of Service 3. Audible Signal Operating Properly	(S) EN	0, 1

		4. Fog Signal Operating Properly		
Lighted Aton Change		1. Light Unlit 2. Light Unreliable 3. Light Re-Establishment 4. Light Range Reduced 5. Light Without Rhythm 6. Light Out Of Synchronization 7. Light Daymark Unreliable 8. Light Operating Properly 9. Sector light Sector obscured 10. Front leading/range light Unlit 11. Rear leading/range light Unlit 12. Front leading/range light Unreliable 13. Rear leading/range light Unreliable 14. Front leading/range light Light range reduced 15. Rear leading/range light Light range reduced 16. Front leading/range light without rhythm 17. Rear leading/range light without rhythm 18. Leading/range lights out of synchronization 19. Front leading/range beacon Unreliable 20. Rear leading/range beacon Unreliable 21. Front leading/range light is operating properly 22. Rear leading/range light is operating properly 23. Front leading/range beacon restored to normal 24. Rear leading/range beacon restored to normal	(S) EN	0, 1

Electronic Aton Change		1. AIS Transmitter Out Of Service 2. AIS Transmitter Unreliable 3. AIS Transmitter Operating Properly 4. V-AIS Out Of Service 5. V-AIS Unreliable 6. V-AIS Operating Properly 7. RACON Out Of Service 8. RACON Unreliable 9. RACON Operating Properly 10. DGPS Out Of Service 11. DGPS Operating Properly 12. DGPS Unreliable 13. LORAN C operating properly 14. LORAN C unreliable 15. LORAN C out of service 16. ELORAN Operating Properly 17. ELORAN Unreliable 18. ELORAN Out Of Service 19. DGLOANSS operating properly 20. DGLOANSS unreliable 21. DGLOANSS out of service 22. Chayka Operating Properly 23. Chayka Unreliable 24. Chayka Out Of Service 25. E-Chayka Operating Properly 26. E-Chayka Unreliable 27. E-Chayka Out Of Service 28. EGNOS Operating Properly 29. EGNOS Unreliable 30. EGNOS Out Of Service	(S) EN	0, 1
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Change Types		1. Advanced notice of changes 2. Discrepancy 3. Proposed changes 4. Temporary changes	EN	0, 1
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5 Simple attributes

5.1 AtoN Number

Type: SimpleAttributeType

Definition: Identifier from a list of Aids to Navigation publication, such as List of Lights.

CamelCase: AtoNNumber

Value type: text

5.2 Category of Aggregation

Type: CodeListType

Definition: Named aggregations between two or more aids to navigation and/or navigationally relevant features.

CamelCase: CategoryOfAggregation

Value type: codelist

Listed values:

Code	Label	Definition
1	leading line	-
3	measured distance	-
2	range system	-

5.3 Category of Association

Type: CodeListType

Definition: Named associations between two or more aids to navigation and/or navigationally relevant features.

CamelCase: CategoryOfAssociation

Value type: codelist

Listed values:

Code	Label	Definition
1	Channel Markings	A group of channel marks which indicate channel limits.
2	Danger Markings	One of more aids to navigation and the danger(s) that are marked.

5.4 Category Of Physical AIS Aid To Navigation

Type: EnumerationType

Definition: A classification of AIS AtoNs that correspond to an actual, physical Aid to Navigation at a real-world location.

CamelCase: CategoryOfPhysicalAISaidToNavigation

Value type: enumeration

Listed values:

Code	Label	Definition
1	Physical AIS Type 1	-
2	Physical AIS Type 2	-
3	Physical AIS Type 3	-

5.5 Category Of Synthetic AIS Aid To Navigation

Type: EnumerationType

Definition: A classification of AIS AtoNs that are transmitted electronically and linked to a real-world object but do not physically exist at the broadcast location.

CamelCase: CategoryOfSyntheticAISaidtoNavigation

Value type: enumeration

Listed values:

Code	Label	Definition
1	predicted	-
2	monitored	-

5.6 Change Types

Type: EnumerationType

Definition: Different categories or kinds of modifications that can be made to data, positions, or objects. For example, changes may involve updates to position, orientation, or attributes of an object.

CamelCase: ChangeTypes

Value type: enumeration

Listed values:

Code	Label	Definition
1	Advanced notice of changes	-
2	Discrepancy	-
3	Proposed changes	-
4	Temporary changes	-

5.7 Seasonal Action Required

Type: SimpleAttributeType

Definition: Indicates whether specific actions or maintenance must be performed based on the season (for example, repositioning, removal, or other adjustments).

CamelCase: SeasonalActionRequired

Value type: text

5.8 Aton Commissioning

Type: EnumerationType

Definition: The process of deploying and activating a new Aid to Navigation (AtoN), ensuring that it is properly installed and operational.

CamelCase: atonCommissioning

Value type: enumeration

Listed values:

Code	Label	Definition
1	Buoy Establishment	A new buoy has been or will be established.
2	Light Establishment	A new light has been or will be established.
3	Beacon Establishment	A new beacon has been or will be established.
4	Audible Signal Establishment	A new audible signal (device activated by e.g. sea state or wind, irrespective of visibility) has been or will be established.
5	Fog Signal Establishment	A new fog signal has been or will be established.
6	AIS Transmitter Establishment	A new AIS site has been or will be established.
7	V-AIS Establishment	A new V-AIS has been or will be established.
8	RACON Establishment	A new RACON has been or will be established.
9	DGPS Station Establishment	A new DGPS station has been or will be established.
10	ELORAN Station Establishment	A new eLORAN station has been or will be established.
11	DGLONASS Station Establishment	A new DGLONASS station has been or will be established.
12	E-Chayka Station Establishment	A new e-Chayka station has been or will be established.
13	EGNOS establishment	.

5.9 Aton Removal

Type: EnumerationType

Definition: The process of decommissioning and physically removing an AtoN from its designated location, either temporarily or permanently.

CamelCase: atonRemoval

Value type: enumeration

Listed values:

Code	Label	Definition
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1	Buoy Removal	Buoy has been or will be permanently removed from service.
2	Buoy Temporary Removal	Buoy has been or will be temporarily removed from service.
3	Light Removal	Light has been or will be permanently removed from service.
4	Light Temporary Removal	Light has been or will be temporarily removed from service.
5	Beacon Removal	Beacon has been or will be permanently removed from service.
6	Beacon Temporary Removal	Beacon has been or will be temporarily removed from service.
7	Fog Signal Removal	Fog signal has been or will be permanently removed from service.
8	Fog Signal Temporary Removal	Fog signal has been or will be temporarily removed from service.
9	Audible Signal Removal	Audible signal (device activated by e.g. sea state or wind, irrespective of visibility) has been or will be permanently removed from service.
10	Audible Signal Temporary Removal	Audible signal (device activated by e.g. sea state or wind, irrespective of visibility) has been or will be temporarily removed from service.
11	V-AIS Removal	V-AIS has been or will be permanently removed from service.
12	V-AIS Temporary Removal	V-AIS has been or will be temporarily removed from service.
13	RACON signal removal	.
14	RACON Temporary Removal	RACON has been or will be temporarily removed from service.
15	DGPS removal	.
16	DGPS temporary removal	.
17	EGNOS removal	.
18	EGNOS temporary removal	.
19	LORAN C Station Removal	LORAN C station has been or will be permanently removed from service.
20	LORAN C Station Temporary Removal	LORAN C station has been or will be temporarily removed from service.
21	eLORAN removal	.
22	eLORAN temporary removal	.
23	Chayka Station Removal	Chayka station has been or will be permanently removed from service.

24	Chayka Station Temporary Removal	Chayka station has been or will be temporarily removed from service.
25	E-Chayka Station Removal	The e-Chayka station has been or will be permanently removed from service.
26	E-Chayka Station Temporary Removal	The e-Chayka station has been or will be temporarily removed from service

5.10 Aton Replacement

Type: EnumerationType

Definition: The act of swapping an existing AtoN with a new or upgraded unit, either due to maintenance needs or technological improvements.

CamelCase: atonReplacement

Value type: enumeration

Listed values:

Code	Label	Definition
1	Buoy Change	The characteristics of the buoy have been or will be changed.
2	Buoy Temporary Change	The characteristics of the buoy have been or will be temporarily changed.
3	Light Change	The characteristics of the light have been or will be changed.
4	Light Temporary Change	The characteristics of the light have been or will be temporarily changed.
5	Sector Light Change	The characteristics of the sector light have been or will be changed.
6	Sector Light Temporary Change	The characteristics of the sector light have been or will be temporarily changed.
7	Beacon Change	The characteristics of the beacon have been or will be changed.
8	Beacon Temporary Change	The characteristics of the beacon have been or will be temporarily changed.
9	Fog Signal Change	The characteristics of the fog signal have been or will be changed.
10	Fog Signal Temporary Change	The characteristics of the fog signal have been or will be temporarily changed.
11	Audible Signal Change	The characteristics of the audible signal (device activated by e.g. sea state or wind, irrespective of visibility) have been or will be changed.
12	Audible Signal Temporary Change	The characteristics of the audible signal (device activated by e.g. sea state or wind, irrespective of visibility) have been or will be temporarily changed.

13	V-AIS Change	The characteristics of the V-AIS have been or will be changed.
14	V-AIS Temporary Change	The characteristics of the V-AIS have been or will be temporarily changed.
15	RACON signal change	.
16	RACON Temporary Change	The characteristics of the RACON have been or will be temporarily changed.

5.11 Audible Signal Aton Change

Type: EnumerationType

Definition: Any modification to an AtoN that uses sound signals, such as foghorns or bells, to assist in navigation under low visibility conditions.

CamelCase: audibleSignalAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	Audible Signal Out Of Service	The audible signal (device activated by e.g. sea state or wind, irrespective of visibility) is inoperative.
2	Fog Signal Out Of Service	The fog signal is inoperative.
3	Audible Signal Operating Properly	The audible signal (device activated by e.g. sea state or wind, irrespective of visibility) is operating as advertised.
4	Fog Signal Operating Properly	The fog signal is operating as advertised.

5.12 Based On Fixed Marks

Type: SimpleAttributeType

Definition: A straight route (known as a recommended track, range or leading line), which comprises: a. at least two structures (usually beacons or daymarks) and/or natural features, which may carry lights and/or top-marks. The structures/features are positioned so that when observed to be in line, a vessel can follow a known bearing with safety. (Adapted from International Association of Lighthouse Authorities - IALA Aids to Navigation Guide, 1990); or b. a single structure or natural feature, which may carry lights and/or a topmark, and a specified bearing which can be followed with safety. (S-57 Edition 3.1, Appendix A Chapter 2, Page 2.72, November 2000, as amended).

CamelCase: basedOnFixedMarks

Value type: boolean

5.13 Beacon Shape

Type: EnumerationType

Definition: Describes the characteristic geometric form of the beacon.

CamelCase: beaconShape

Value type: enumeration

Listed values:

Code	Label	Definition
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1	Stake, Pole, Perch, Post	An elongated wood or metal pole, driven into the ground or seabed, which serves as a navigational aid or a support for a navigational aid.
2	Withy	A tree without roots stuck or spoiled into the bottom of the sea to serve as a navigational aid.
3	Beacon Tower	A solid structure of the order of 10 metres in height used as a navigational aid.
4	Lattice Beacon	A structure consisting of strips of metal or wood crossed or interlaced to form a structure to serve as an aid to navigation or as a support for an aid to navigation.
5	Pile Beacon	A long heavy timber(s) or section(s) of steel, wood, concrete, etc., forced into the seabed to serve as an aid to navigation or as a support for an aid to navigation.
6	Cairn	A mound of stones, usually conical or pyramidal, raised as a landmark or to designate a point of importance in surveying.
7	Buoyant Beacon	A tall spar-like beacon fitted with a permanently submerged buoyancy chamber, the lower end of the body is secured to seabed sinker either by a flexible joint or by a cable under tension.

5.14 Building Shape

Type: EnumerationType

Definition: The specific shape of the building.

CamelCase: buildingShape

Value type: enumeration

Listed values:

Code	Label	Definition
5	High-Rise Building	A building having many storeys.
6	Pyramid	A polyhedron of which one face is a polygon of any number of sides, and the other faces are triangles with a common vertex.
7	Cylindrical	Shaped like a cylinder, which is a solid geometrical figure generated by straight lines fixed in direction and describing with one of its points a closed curve, especially a circle.
8	Spherical	Shaped like a sphere, which is a body the surface of which is at all points equidistant from the centre.
9	Cubic	A shape the sides of which are six equal squares; a regular hexahedron.

5.15 Buoy Shape

Type: EnumerationType

Definition: The principal shape and/or design of a buoy.

CamelCase: buoyShape

Value type: enumeration

Listed values:

Code	Label	Definition
1	Conical	The upper part of the body above the water-line, or the greater part of the superstructure, has approximately the shape or the appearance of a pointed cone with the point upwards.
2	Can	The upper part of the body above the water-line, or the greater part of the superstructure, has the shape of a cylinder, or a truncated cone that approximates to a cylinder, with a flat end uppermost.
3	Spherical	Shaped like a sphere, which is a body the surface of which is at all points equidistant from the centre.
4	Pillar	The upper part of the body above the water-line, or the greater part of the superstructure is a narrow vertical structure, pillar or lattice tower.
5	Spar	The upper part of the body above the water-line, or the greater part of the superstructure, has the form of a pole, or of a very long cylinder, floating upright.
6	Barrel	The upper part of the body above the water-line, or the greater part of the superstructure, has the form of a barrel or cylinder floating horizontally.
7	Superbuoy	A very large buoy designed to carry a signal light of high luminous intensity at a high elevation.
8	Ice Buoy	A specially constructed shuttle shaped buoy which is used in ice conditions.

5.16 Category of Cardinal Mark

Type: EnumerationType

Definition: The four quadrants (north, east, south and west) are bounded by the true bearings NW-NE, NE-SE, SE-SW and SW-NW taken from the point of interest. A cardinal mark is named after the quadrant in which it is placed. The name of the cardinal mark indicates that it should be passed to the named side of the mark.

CamelCase: categoryOfCardinalMark

Value type: enumeration

Listed values:

Code	Label	Definition
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1	North Cardinal Mark	Quadrant bounded by the true bearing NW-NE taken from the point of interest; it should be passed to the north side of the mark.
2	East Cardinal Mark	Quadrant bounded by the true bearing NE-SE taken from the point of interest. It should be passed to the east side of the mark.
3	South Cardinal Mark	Quadrant bounded by the true bearing SE-SW taken from the point of interest; it should be passed to the south side of the mark.
4	West Cardinal Mark	Quadrant bounded by the true bearing SW-NW taken from the point of interest; it should be passed to the west side of the mark.

5.17 Category of Fog Signal

Type: EnumerationType

Definition: Classification of the various means of generating the fog signal.

CamelCase: categoryOfFogSignal

Value type: enumeration

Listed values:

Code	Label	Definition
1	Explosive	A signal produced by the firing of explosive charges.
2	Diaphone	A diaphone uses compressed air and generally emits a powerful low-pitched sound, which often concludes with a brief sound of suddenly lowered pitch, termed the 'grunt'.
3	Siren	A type of fog signal apparatus which produces sound by virtue of the passage of air through slots or holes in a revolving disk.
4	Nautophone	A horn having a diaphragm oscillated by electricity.
5	Reed	[1] A reed uses compressed air and emits a weak, high pitched sound. [2] Any of various water or marsh plants with a firm stem. (Concise Oxford English Dictionary)
6	Tyfon	A diaphragm horn which operates under the influence of compressed air or steam.
7	Bell	A ringing sound with a short range.
8	Whistle	A distinctive sound made by a jet of air passing through an orifice. The apparatus may be operated automatically, by hand or by air being forced up a tube by waves acting on a buoy.

9	Gong	A sound produced by vibration of a disc when struck.
10	Horn	A horn uses compressed air or electricity to vibrate a diaphragm and exists in a variety of types which differ greatly in their sound and power.

5.18 Category of Installation Buoy

Type: EnumerationType

Definition: Classification of fixed installation buoy.

CamelCase: categoryOfInstallationBuoy

Value type: enumeration

Listed values:

Code	Label	Definition
1	Catenary Anchor Leg Mooring	Incorporates a large buoy which remains on the surface at all times and is moored by 4 or more anchors. Mooring hawsers and cargo hoses lead from a turntable on top of the buoy, so that the buoy does not turn as the ship swings to wind and stream.
2	Single Buoy Mooring	A large mooring buoy used by tankers to load and unload in port approaches or in offshore oil and gas fields.

5.19 Category of Landmark

Type: EnumerationType

Definition: Classification of prominent cultural and natural features in the landscape.

CamelCase: categoryOfLandmark

Value type: enumeration

Listed values:

Code	Label	Definition
1	Cairn	A mound of stones, usually conical or pyramidal, raised as a landmark or to designate a point of importance in surveying.
2	Cemetery	A site and associated structures devoted to the burial of the dead.
3	Chimney	A vertical structure containing a passage or flue for discharging smoke and gases of combustion.
4	Dish Aerial	A parabolic aerial for the receipt and transmission of high frequency radio signals.
5	Flagstaff	A staff or pole on which flags are raised.
6	Flare Stack	A tall structure used for burning-off waste oil or gas.

7	Mast	A relatively tall structure usually held vertical by guy lines.
8	Windsock	A tapered fabric sleeve mounted so as to catch and swing with the wind, thus indicating the wind direction.
9	Monument	A structure erected and/or maintained as a memorial to a person and/or event.
10	Column/Pillar	A cylindrical or slightly tapering body of considerably greater length than diameter erected vertically.
11	Memorial Plaque	A slab of metal, usually ornamented, erected as a memorial to a person or event.
12	Obelisk	A tapering shaft usually of stone or concrete, square or rectangular in section, with a pyramidal apex.
13	Statue	A representation of a living being, sculptured, moulded, or cast in a variety of materials (for example: marble, metal, or plaster).
14	Cross	A monument, or other structure in form of a cross.
15	Dome	A landmark comprising a hemispherical or spheroidal shaped structure.
16	Radar Scanner	A device used for directing a radar beam through a search pattern.
17	Tower	A relatively tall, narrow structure that may either stand alone or may form part of another structure.
18	Windmill	A system of vanes attached to a tower and driven by wind (excluding wind turbines).
19	Windmotor	A modern structure for the use of wind power.
20	Spire/Minaret	A tall conical or pyramid-shaped structure often built on the roof or tower of a building, especially a church or mosque.
21	Large Rock or Boulder on Land	An isolated rocky formation or a single large stone.
22	Triangulation Mark	A recoverable point on the earth, whose geographic position has been determined by angular methods with geodetic instruments. A triangulation point is a selected point, which has been marked with a station mark, or it is a conspicuous natural or artificial feature.
23	Boundary Mark	A marker identifying the location of a surveyed boundary line.
24	Observation Wheel	Wheels with passenger cars mounted external to the rim and independently

		rotated by electric motors.
25	Torii	A form of decorative gateway or portal, consisting of two upright wooden posts connected at the top by two horizontal crosspieces, commonly found at the entrance to Shinto temples.
26	Bridge	A structure erected over a depression or an obstacle such as a body of water, railroad, etc., to provide a roadway for vehicles or pedestrians.
27	Dam	A barrier to check or confine anything in motion; particularly one constructed to hold back water and raise its level to form a reservoir, or to prevent flooding.

5.20 Category of Lateral Mark

Type: EnumerationType

Definition: Classification of lateral marks in the IALA Buoyage System.

CamelCase: categoryOfLateralMark

Value type: enumeration

Listed values:

Code	Label	Definition
1	Port-Hand Lateral Mark	Indicates the port boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
2	Starboard-Hand Lateral Mark	Indicates the starboard boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
3	Preferred Channel to Starboard Lateral Mark	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified port-hand lateral mark.
4	Preferred Channel to Port Lateral Mark	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark.
5	Right-Hand Side of the Waterway	Indicates the right-hand side of the inland waterway.
6	Left-Hand Side of the Waterway	Indicates the left-hand side of the inland waterway.
7	Right-Hand Side of the Channel	Indicates the right-hand side of a channel of an inland waterway.
8	Left-Hand Side of the Channel	Indicates the left-hand side of a channel of an inland waterway.
9	Bifurcation of the Waterway	Indicates a bifurcation of the inland waterway.

10	Bifurcation of the Channel	Indicates a bifurcation of a channel of an inland waterway.
11	Channel Near the Right Bank	Indicates that the channel is near the right bank.
12	Channel Near the Left Bank	Indicates that the channel is near the left bank.
13	Channel Cross-Over to the Right Bank	Indicates that the channel crosses from the left to the right bank.
14	Channel Cross-Over to the Left Bank	Indicates that the channel crosses from the right to the left bank.
15	Danger Point or Obstacles at the Right-Hand Side	Indicates a danger point or obstacles at the right-hand side.
16	Danger Point or Obstacles at the Left-Hand Side	Indicates a danger point or obstacles at the left-hand side.
17	Turn Off at the Right-Hand Side	Indicates a turn off at the right-hand side.
18	Turn Off at the Left-Hand Side	Indicates a turn off at the left-hand side.
19	Junction at the Right-Hand Side	Indicates a junction at the right-hand side.
20	Junction at the Left-Hand Side	Indicates a junction at the left-hand side.
21	Harbour Entry at the Right-Hand Side	Indicates a harbour entry at the right-hand side.
22	Harbour Entry at the Left-Hand Side	Indicates a harbour entry at the left-hand side.
23	Bridge Pier Mark	Indicates a bridge pier in an inland waterway.
24	Entry From a Lake to a Narrower Waterway, Right Bank	Indicates the right bank of the entry from a lake or a lake-like expansion to a section of the waterway which is narrower.
25	Entry From a Lake to a Narrower Waterway, Left Bank	Indicates the left bank of the entry from a lake or a lake-like expansion to a section of the waterway which is narrower.
26	Change Bank	Change bank.
27	Continue Along Bank	Continue along bank.

5.21 Category of Light

Type: EnumerationType

Definition: Classification of different light types.

CamelCase: categoryOfLight

Value type: enumeration

Listed values:

Code	Label	Definition
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1	Directional Function	A light illuminating a sector of very narrow angle and intended to mark a direction to follow.
4	Leading Light	A light associated with other lights so as to form a leading line to be followed.
5	Aero Light	An aero light is established for aeronautical navigation and may be of higher power than marine lights and visible from well offshore.
6	Air Obstruction Light	A light marking an obstacle which constitutes a danger to air navigation.
8	Flood Light	A broad beam light used to illuminate a structure or area.
9	Strip Light	A light whose source has a linear form generally horizontal, which can reach a length of several metres.
10	Subsidiary Light	A light placed on or near the support of a main light and having a special use in navigation.
11	Spotlight	A powerful light focused so as to illuminate a small area.
12	Front	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
13	Rear	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
14	Lower	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
15	Upper	Term used with leading lights to describe the position of the light on the lead as viewed from seaward.
17	Emergency	A light available as a backup to a main light which will be illuminated should the main light fail.
18	Bearing Light	A light which enables its approximate bearing to be obtained without the use of a compass.
19	Horizontally Disposed	A group of lights of identical character and almost identical position, that are disposed horizontally.
20	Vertically Disposed	A group of lights of identical character and almost identical position, that are disposed vertically.

5.22 Category of Navigation Line

Type: EnumerationType

Definition: Classification of route guidance given to vessels.

CamelCase: categoryOfNavigationLine

Value type: enumeration

Listed values:

Code	Label	Definition
1	Clearing Line	A straight line that marks the boundary between a safe and a dangerous area or that passes clear of a navigational danger.
2	Transit Line	A line passing through one or more fixed marks.
3	Leading Line Bearing a Recommended Track	A line passing through one or more clearly defined objects, along the path of which a vessel can approach safely up to a certain distance off.

5.23 Category of Offshore Platform

Type: EnumerationType

Definition: Classification of an offshore raised structure.

CamelCase: categoryOfOffshorePlatform

Value type: enumeration

Listed values:

Code	Label	Definition
1	Oil Rig	A temporary mobile structure, either fixed or floating, used in the exploration stages of oil and gas fields.
2	Production Platform	A term used to indicate a permanent offshore structure equipped to control the flow of oil or gas. It does not include entirely submarine structures.
3	Observation/Research Platform	A platform from which one's surroundings or events can be observed, noted or recorded such as for scientific study.
4	Articulated Loading Platform	A metal lattice tower, buoyant at one end and attached at the other by a universal joint to a concrete filled base on the sea bed. The platform may be fitted with a helicopter platform, emergency accommodation and hawser/hose retrieval.
5	Single Anchor Leg Mooring	A rigid frame or tube with a buoyancy device at its upper end , secured at its lower end to a universal joint on a large steel or concrete base resting on the sea bed, and at its upper end to a mooring buoy by a chain or wire.
6	Mooring Tower	A platform secured to the sea bed and surmounted by a turntable to which ships moor.
7	Artificial Island	A man-made structure usually built for the exploration or exploitation of marine

		resources, marine scientific research, tidal observations, etc.
8	Floating Production, Storage and Off-Loading Vessel	An offshore oil/gas facility consisting of a moored tanker/barge by which the product is extracted, stored and exported.
9	Accommodation Platform	A platform used primarily for eating, sleeping and recreation purposes.
10	Navigation, Communication and Control Buoy	A floating structure with control room, power and storage facilities, attached to the sea bed by a flexible pipeline and cables.
11	Floating Oil Tank	A floating structure, anchored to the seabed, for storing oil.

5.24 Category of Pile

Type: EnumerationType

Definition: Classification of pile, driven into the earth as a foundation or support for a structure.

CamelCase: categoryOfPile

Value type: enumeration

Listed values:

Code	Label	Definition
1	Stake	An elongated wood or metal pole embedded in the seabed to serve as a marker or support.
3	Post	A vertical piece of timber, metal or concrete forced into the earth or sea bed.
4	Tripodal	A single structure comprising 3 or more piles held together (sections of heavy timber, steel or concrete), and forced into the earth or sea bed.
5	Piling	A number of piles, usually in a straight line, and usually connected or bolted together.
6	Area of Piles	A number of piles, usually in a straight line, but not connected by structural members.
7	Pipe	A vertical hollow cylinder of metal, wood, or other material forced into the earth or seabed.
8	Mooring Post	A post where to which something (such as a craft) can be moored.

5.25 Category of Radar Transponder Beacon

Type: EnumerationType

Definition: Classification of radar transponder beacon based on functionality.

CamelCase: categoryOfRadarTransponderBeacon

Value type: enumeration

Listed values:

Code	Label	Definition
1	Ramark, Radar Beacon Transmitting Continuously	A radar marker beacon which continuously transmits a signal appearing as a radial line on a radar screen, the line indicating the direction of the beacon. Ramarks are intended primarily for marine use. The name 'ramark' is derived from the words radar marker.
2	Racon, Radar Transponder Beacon	A radar beacon which returns a coded signal which provides identification of the beacon, as well as range and bearing. The range and bearing are indicated by the location of the first character received on the radar screen. The name 'racon' is derived from the words radar beacon.
3	Leading Racon/Radar Transponder Beacon	A radar beacon that may be used (in conjunction with at least one other radar beacon) to indicate a leading line.

5.26 Category of Radio Station

Type: EnumerationType

Definition: Classification of radio services offered by a radio station.

CamelCase: categoryOfRadioStation

Value type: enumeration

Listed values:

Code	Label	Definition
1	Circular (Non-Directional) Marine or Aero-Marine Radiobeacon	A radio station which need not necessarily be manned, the emissions of which, radiated around the horizon, enable its bearing to be determined by means of the radio direction finder of a ship.
2	Directional Radiobeacon	A special type of radiobeacon station the emissions of which are intended to provide a definite track for guidance.
3	Rotating Pattern Radiobeacon	A special type of radiobeacon station emitting a beam of waves to which a uniform turning movement is given, the bearing of the station being determined by means of an ordinary listening receiver and a stop watch. Also referred to as a rotating loop radiobeacon.
4	Consol Beacon	A type of long range position fixing beacon.
5	Radio Direction-Finding Station	A radio station intended to determine only the direction of other stations by means of transmission from the latter.
6	Coast Radio Station Providing QTG Service	A radio station which is prepared to provide QTG service; that is to say, to transmit upon request from a ship a

		radio signal, the bearing of which can be taken by that ship.
7	Aeronautical Radiobeacon	A radio beacon designed for aeronautical use.
8	Decca	The Decca Navigator System is a high accuracy, short to medium range radio navigational aid intended for coastal and landfall navigation.
9	Loran C	A low frequency electronic position fixing system using pulsed transmissions at 100 Khz.
10	Differential GNSS	A radiobeacon transmitting DGPS correction signals.
11	Toran	An electronic position fixing system used mainly by aircraft.
12	Omega	A long-range radio navigational aid which operates within the VLF frequency band. The system comprises eight land based stations.
13	Syledis	A ranging position fixing system operating at 420-450 MHz over a range of up to 400 Km.
14	Chaika	Chaika is a low frequency electronic position fixing system using pulsed transmissions at 100 Khz.
19	Radio Telephone Station	The equipment needed at one station to carry on two way voice communication by radio waves only.
20	AIS Base Station	An onshore AIS unit that monitors traffic in the waterways.

5.27 Category of Silo/Tank

Type: EnumerationType

Definition: Classification based on the product for which a silo or tank is used.

CamelCase: categoryOfSiloTank

Value type: enumeration

Listed values:

Code	Label	Definition
1	Silo in General	A large storage structure used for storing loose materials.
2	Tank in General	A fixed structure for storing liquids.
3	Grain Elevator	A storage building for grain. Usually a tall frame, metal or concrete structure with an especially compartmented interior.
4	Water Tower	A tower supporting an elevated storage tank of water.

5.28 Category of Special Purpose Mark

Type: EnumerationType

Definition: Classification of an aid to navigation which signifies some special purpose.

CamelCase: categoryOfSpecialPurposeMark

Value type: enumeration

Listed values:

Code	Label	Definition
1	Firing Danger Area Mark	A mark used to indicate a firing danger area, usually at sea.
2	Target Mark	Any object toward which something is directed. The distinctive marking or instrumentation of a ground point to aid its identification on a photograph.
3	Marker Ship Mark	A mark marking the position of a ship which is used as a target during some military exercise.
4	Degaussing Range Mark	A mark used to indicate a degaussing range.
5	Barge Mark	A mark of relevance to barges.
6	Cable Mark	A mark used to indicate the position of submarine cables or the point at which they run on to the land.
7	Spoil Ground Mark	A mark used to indicate the limit of a spoil ground.
8	Outfall Mark	A mark used to indicate the position of an outfall or the point at which it leaves the land.
9	ODAS	Ocean Data Acquisition System.
10	Recording Mark	A mark used to record data for scientific purposes.
11	Seaplane Anchorage Mark	A mark used to indicate a seaplane anchorage.
12	Recreation Zone Mark	A mark used to indicate a recreation zone.
13	Private Mark	A privately maintained mark.
14	Mooring Mark	A mark indicating a mooring or moorings.
15	LANBY	A large buoy designed to take the place of a lightship where construction of an offshore light station is not feasible.
16	Leading Mark	Aids to navigation or other indicators so located as to indicate the path to be followed. Leading marks identify a leading line when they are in transit.

17	Measured Distance Mark	A mark forming part of a transit indicating one end of a measured distance.
18	Notice Mark	A notice board or sign indicating information to the mariner.
19	TSS Mark	A mark indicating a Traffic Separation Scheme.
20	Anchoring Prohibited Mark	A mark indicating an anchoring prohibited area.
21	Berthing Prohibited Mark	A mark indicating that berthing is prohibited.
22	Overtaking Prohibited Mark	A mark indicating that overtaking is prohibited.
23	Two-Way Traffic Prohibited Mark	A mark indicating a one-way route.
24	Reduced Wake Mark	A mark indicating that vessels must not generate excessive wake.
25	Speed Limit Mark	A mark indicating that a speed limit applies.
26	Stop Mark	A mark indicating the place where the bow of a ship must stop when traffic lights show red.
27	General Warning Mark	A mark indicating that special caution must be exercised in the vicinity of the mark.
28	Sound Ship's Siren Mark	A mark indicating that a ship should sound its siren or horn.
29	Restricted Vertical Clearance Mark	A mark indicating the minimum vertical space available for passage.
30	Maximum Vessel's Draught Mark	A mark indicating the maximum draught of vessel permitted.
31	Restricted Horizontal Clearance Mark	A mark indicating the minimum horizontal space available for passage.
32	Strong Current Warning Mark	A mark warning of strong currents.
33	Berthing Permitted Mark	A mark indicating that berthing is allowed.
34	Overhead Power Cable Mark	A mark indicating an overhead power cable.
35	Channel Edge Gradient Mark	A mark indicating the gradient of the slope of a dredge channel edge.
36	Telephone Mark	A mark indicating the presence of a telephone.
37	Ferry Crossing Mark	A mark indicating that a ferry route crosses the ship route; often used with a 'sound ship's siren' mark.

39	Pipeline Mark	A mark used to indicate the position of submarine pipelines or the point at which they run on to the land.
40	Anchorage Mark	A mark indicating an anchorage area.
41	Clearing Mark	A mark used to indicate a clearing line.
42	Control Mark	A mark indicating the location at which a restriction or requirement exists.
43	Diving Mark	A mark indicating that diving may take place in the vicinity.
44	Refuge Beacon	A mark providing or indicating a place of safety.
45	Foul Ground Mark	A mark indicating a foul ground.
46	Yachting Mark	A mark installed for use by yachtsmen.
47	Heliport Mark	A mark indicating an area where helicopters may land.
48	GNSS Mark	A mark indicating a location at which a GNSS position has been accurately determined.
49	Seaplane Landing Mark	A mark indicating an area where sea-planes land.
50	Entry Prohibited Mark	A mark indicating that entry is prohibited.
51	Work in Progress Mark	A mark indicating that work (generally construction) is in progress.
52	Mark With Unknown Purpose	A mark whose detailed characteristics are unknown.
53	Wellhead Mark	A mark indicating a borehole that produces or is capable of producing oil or natural gas.
54	Channel Separation Mark	A mark indicating the point at which a channel divides separately into two channels.
55	Marine Farm Mark	A mark indicating the existence of a fish, mussel, oyster or pearl farm/culture.
56	Artificial Reef Mark	A mark indicating the existence or the extent of an artificial reef.
57	Ice Mark	A mark, used year round, that may be submerged when ice passes through the area.
58	Nature Reserve Mark	A mark used to define the boundary of a nature reserve.
59	Fish Aggregating Device	A fish aggregating (or aggregation) device (FAD) is a man-made object used to attract ocean going pelagic fish such as marlin, tuna and mahi-mahi (dolphin fish). They usually consist of buoys or

		floats tethered to the ocean floor with concrete blocks.
60	Wreck Mark	A mark used to indicate the existence of a wreck.
61	Customs Mark	A mark used to indicate the existence of a customs checkpoint.
62	Causeway Mark	A mark used to indicate the existence of a causeway.
63	Wave Recorder	A surface following buoy used to measure wave activity.
64	Jetski Prohibited	A mark indicating a jetski prohibited area.

5.29 Colour

Type: EnumerationType

Definition: The property possessed by an object of producing different sensations on the eye as a result of the way it reflects or emits light.

CamelCase: colour

Value type: enumeration

Listed values:

Code	Label	Definition
1	White	The achromatic object colour of greatest lightness characteristically perceived to belong to objects that reflect diffusely nearly all incident energy throughout the visible spectrum.
2	Black	The achromatic color of least lightness characteristically perceived to belong to objects that neither reflect nor transmit light.
3	Red	A color whose hue resembles that of blood or of the ruby or is that of the long-wave extreme of the visible spectrum.
4	Green	Of the color green.
5	Blue	A color whose hue is that of the clear sky or that of the portion of the color spectrum lying between green and violet.
6	Yellow	A color whose hue resembles that of ripe lemons or sunflowers or is that of the portion of the spectrum lying between green and orange.
7	Grey	Of the color grey.
8	Brown	Any of a group of colors between red and yellow in hue, of medium to low lightness, and of moderate to low saturation.
9	Amber	A variable color averaging a dark orange

		yellow.
10	Violet	Any of a group of colors of reddish-blue hue, low lightness, and medium saturation.
11	Orange	Any of a group of colors that are between red and yellow in hue.
12	Magenta	A deep purplish red.
13	Pink	Any of a group of colors bluish red to red in hue, of medium to high lightness, and of low to moderate saturation.

5.30 Colour Pattern

Type: EnumerationType

Definition: A regular repeated design containing more than one colour.

CamelCase: colourPattern

Value type: enumeration

Listed values:

Code	Label	Definition
1	Horizontal Stripes	Straight bands or stripes of differing colours oriented horizontally.
2	Vertical Stripes	Straight bands or stripes of differing colours oriented vertically.
3	Diagonal Stripes	Straight bands or stripes of differing colours oriented diagonally (that is, not horizontally or vertically).
4	Squared	Often referred to as checker plate, where alternate colours are used to create squares similar to a chess or draught board. The pattern may be straight or diagonal.
5	Stripes (Direction Unknown)	Straight bands or stripes of differing colours oriented in an unknown direction.
6	Border Stripe	A band or stripe of colour which is displayed around the outer edge of the object, which may also form a border to an inner pattern or plain colour.
7	Single Colour	One solid colour of uniform coverage.
8	Rectangle	A four-sided shape that is made up of two pairs of parallel lines and that has four right angles, on a different coloured background.
9	Triangle	A shape that is made up of three lines and three angles, on a different coloured background.

5.31 Condition

Type: EnumerationType

Definition: The various conditions of buildings and other constructions.

CamelCase: condition

Value type: enumeration

Listed values:

Code	Label	Definition
1	Under Construction	Being built but not yet capable of function.
2	Ruined	A structure in a decayed or deteriorated condition resulting from neglect or disuse, or a damaged structure in need of repair.
3	Under Reclamation	An area of the sea, a lake or the navigable part of a river that is being reclaimed as land, usually by the dumping of earth and other material.
4	Wingless	A windmill or wind turbine from which the vanes or turbine blades are missing.
5	Planned Construction	Detailed planning has been completed but construction has not been initiated.

5.32 Date End

Type: SimpleAttributeType

Definition: The latest date on which an object (for example a buoy) will be present.

CamelCase: dateEnd

Value type: S100_TruncatedDate

5.33 Date Start

Type: SimpleAttributeType

Definition: The earliest date on which an object (for example a buoy) will be present.

CamelCase: dateStart

Value type: S100_TruncatedDate

5.34 Depth Range Minimum Value

Type: SimpleAttributeType

Definition: The minimum (shoalest) value of a depth range.

CamelCase: depthRangeMinimumValue

Value type: real

5.35 Effective Intensity

Type: SimpleAttributeType

Definition: The luminous intensity of a fictitious juxtaposed steady-burning point light source that would appear to exhibit a luminosity equal to that of the rhythmic point light source it describes. The apparent reduction in intensity of the rhythmic light is subjective and is due to the nature of the response of the eye of the observer.

CamelCase: effectiveIntensity

Value type: real

5.36 Electronic Aton Change

Type: EnumerationType

Definition: Modifications to electronic or digital AtoNs, such as AIS (Automatic Identification System) AtoNs, virtual AtoNs, or remote-controlled systems.

CamelCase: electronicAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	AIS Transmitter Out Of Service	The terrestrial AIS transmitter is inoperative due to a technical issue.
2	AIS Transmitter Unreliable	The terrestrial AIS transmitter is unreliable due to a technical issue or maintenance.
3	AIS Transmitter Operating Properly	The terrestrial AIS transmitter is operating as advertised.
4	V-AIS Out Of Service	Virtual AIS aid to navigation is extinguished.
5	V-AIS Unreliable	Virtual AIS aid is unreliable due to a technical issue or maintenance.
6	V-AIS Operating Properly	Virtual AIS aid to navigation is operating as advertised.
7	RACON Out Of Service	The RACON is inoperative.
8	RACON Unreliable	The RACON is unreliable due to a technical issue or maintenance.
9	RACON Operating Properly	The RACON is operating as advertised.
10	DGPS Out Of Service	The DGPS station is inoperative due to a technical issue.
11	DGPS Operating Properly	The DGPS station is operating as advertised.
12	DGPS Unreliable	The DGPS station is unreliable due to a technical issue or maintenance.
13	LORAN C operating properly	.
14	LORAN C unreliable	.
15	LORAN C out of service	.
16	ELORAN Operating Properly	The eLORAN station is operating as advertised.
17	ELORAN Unreliable	The eLORAN station is unreliable due to a technical issue or maintenance.
18	ELORAN Out Of Service	The eLORAN station is inoperative due to a technical issue.
19	DGLOANSS operating properly	.
20	DGLOANSS unreliable	.

21	DGLOANSS out of service	.
22	Chayka Operating Properly	The Chayka station is operating as advertised.
23	Chayka Unreliable	The Chayka station is unreliable due to a technical issue or maintenance.
24	Chayka Out Of Service	The Chayka station is inoperative due to a technical issue.
25	E-Chayka Operating Properly	The e-Chayka station is operating as advertised.
26	E-Chayka Unreliable	The e-Chayka station is unreliable due to a technical issue or maintenance.
27	E-Chayka Out Of Service	The e-Chayka station is inoperative due to a technical issue.
28	EGNOS Operating Properly	The EGNOS station is operating as advertised.
29	EGNOS Unreliable	The EGNOS station is unreliable due to a technical issue or maintenance.
30	EGNOS Out Of Service	The EGNOS station is inoperative due to a technical issue.

5.37 Estimated Range of Transmission

Type: SimpleAttributeType

Definition: The estimated range of a non-optical electromagnetic transmission.

CamelCase: estimatedRangeOfTransmission

Value type: real

5.38 Exhibition Condition of Light

Type: EnumerationType

Definition: The outward display of the light.

CamelCase: exhibitionConditionOfLight

Value type: enumeration

Listed values:

Code	Label	Definition
1	Light Shown Without Change of Character	A light shown throughout the 24 hours without change of character.
2	Daytime Light	A light which is only exhibited by day.
3	Fog Light	A light which is exhibited in fog or conditions of reduced visibility.
4	Night Light	A light which is only exhibited at night.

5.39 File Locator

Type: SimpleAttributeType

Definition: The location of a fragment of text or other information in a support file.

CamelCase: fileLocator

Value type: text

5.40 File Reference

Type: SimpleAttributeType

Definition: The file name of an externally referenced text file.

CamelCase: fileReference

Value type: text

5.41 Fixed Aton Change

Type: EnumerationType

Definition: Modifications or updates to fixed AtoNs, such as lighthouses or beacons, which are permanently positioned.

CamelCase: fixedAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	Beacon Missing	No beacon at the advertised position.
2	Beacon Damaged	The beacon has sustained damage due to external factors (wind, sea state, collision with a vessel).
3	Light beacon Unlit	.
4	Light beacon Unreliable	.
5	Light beacon Not synchronized	.
6	Light beacon damaged	.
7	Beacon Topmark Missing	The topmark of the beacon is missing.
8	Beacon Topmark Damaged	The topmark of the beacon is damaged due to external factors (wind, sea state, collision with a vessel).
9	Beacon Daymark Unreliable	Colour of the beacon daymark is not visible due to damage or fading of colours (out of tolerance with colour recommendations).
10	Floodlit beacon Unlit	.
11	Beacon Restored To Normal	The beacon has been restored to normal condition.

5.42 Flare Bearing

Type: SimpleAttributeType

Definition: The bearing about which the light flare symbol is rotated to be displayed in ECDIS.

CamelCase: flareBearing

Value type: integer

5.43 Floating Aton Change

Type: EnumerationType

Definition: Adjustments or replacements related to floating AtoNs, such as buoys, which are anchored but can move with water currents.

CamelCase: floatingAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	Buoy Adrift	The buoy is no longer secured to its moorings and is adrift.
2	Buoy Damaged	The buoy has been damaged due to external factors (wind, sea state, collision with a vessel).
3	Buoy Daymark Unreliable	Colour of the buoy daymark is not visible due to damage or fading of colours (out of tolerance with colour recommendations).
4	Buoy Destroyed	The buoy has suffered extensive damage and is not useable.
5	Buoy Missing	No buoy at its advertised/charted position or in the vicinity.
6	Buoy Move	The buoy has been or will be moved intentionally.
7	Buoy off Position	The buoy has been dragged off its advertised position due to wind or current affecting the mooring system.
8	Buoy re-establishment	.
9	Buoy Restored to Normal	The buoy has been restored to normal condition.
10	Buoy Topmark Damaged	The topmark of the buoy is damaged due to external factors (wind, sea state, collision with a vessel).
11	Buoy Topmark Missing	The topmark of the buoy is missing.
12	Buoy Will Be Withdrawn	The buoy has been scheduled for removal from service for a fixed term.
13	Buoy Withdrawn	The buoy has been removed from service for a fixed term.
14	Decommissioned for winter	.
15	Lifted for Winter	.
16	Light buoy Light damaged	.
17	Light buoy Light not synchronized	.
18	Light buoy Light unlit	.

19	Light buoy Light unreliable	.
20	Marine Aids to Navigation Unreliable	The position or status of Marine Aids to Navigation, over an extensive area, is unreliable due to a natural event (freshet, storm surge, flooding).
21	Recommissioned for navigation season	.
22	Replaced by Winter Spar	.
23	Seasonal Decommissioning Complete	The completion of the process to remove summer buoys (and possibly replace some with winter spar buoys).
24	Seasonal Decommissioning in Progress	The commencement of the process to remove summer buoys (and possibly replace some with winter spar buoys).
25	Seasonal recommissioning complete	.
26	Seasonal recommissioning in progress	.

5.44 Function

Type: EnumerationType

Definition: A specific role that describes a feature.

CamelCase: functionFihoXyz

Value type: enumeration

Listed values:

Code	Label	Definition
2	Harbour-Masters Office	A local official who has charge of mooring and berthing of vessels, collecting harbour fees, etc.
3	Customs Office	Serves as a government office where customs duties are collected, the flow of goods are regulated and restrictions enforced, and shipments or vehicles are cleared for entering or leaving a country.
4	Health Office	The office which is charged with the administration of health laws and sanitary inspections.
5	Hospital	An institution or establishment providing medical or surgical treatment for the ill or wounded.
6	Post Office	The public department, agency or organisation responsible primarily for the collection, transmission and distribution of mail.
7	Hotel	An establishment, especially of a comfortable or luxurious kind, where paying visitors are provided with accommodation, meals and other services.
8	Railway Station	A building with platforms where trains

		arrive, load, discharge and depart.
9	Police Station	The headquarters of a local police force and that is where those under arrest are first charged.
10	Water-Police Station	The headquarters of a local water-police force.
11	Pilot Office	The office or headquarters of pilots; the place where the services of a pilot may be obtained.
12	Pilot Lookout	A distinctive structure or place on shore from which personnel keep watch upon events at sea or along the coast.
13	Bank Office	An office for custody, deposit, loan, exchange or issue of money.
14	Headquarters for District Control	The quarters of an executive officer (director, manager, etc.) with responsibility for an administrative area.
15	Transit Shed/Warehouse	A building or part of a building for storage of wares or goods.
16	Factory	A building or buildings with equipment for manufacturing; a workshop.
17	Power Station	A stationary plant containing apparatus for large scale conversion of some form of energy (such as hydraulic, steam, chemical or nuclear energy) into electrical energy.
18	Administrative	A building for the management of affairs.
19	Educational Facility	A building concerned with education (for example school, college, university, etc).
20	Church	A building for public Christian worship.
21	Chapel	A place for Christian worship other than a parish, cathedral or church, especially one attached to a private house or institution.
22	Temple	A building for public Jewish worship.
23	Pagoda	A Hindu or Buddhist temple or sacred building.
24	Shinto Shrine	A building for public Shinto worship.
25	Buddhist Temple	A building for public Buddhist worship.
26	Mosque	A Muslim place of worship.
27	Marabout	A shrine marking the burial place of a Muslim holy man.
28	Lookout	Keeping a watch upon events at sea or along the coast.
29	Communication	Transmitting and/or receiving electronic

		communication signals.
30	Television	A system for reproducing on a screen visual images transmitted (usually with sound) by radio signals.
31	Radio	Transmitting and/or receiving radio-frequency electromagnetic waves as a means of communication.
32	Radar	A method, system or technique of using beamed, reflected, and timed radio waves for detecting, locating, or tracking objects, and for measuring altitudes.
33	Light Support	A structure serving as a support for one or more lights.
34	Microwave	Broadcasting and receiving signals using microwaves.
35	Cooling	Generation of chilled liquid and/or gas for cooling purposes.
36	Observation	A place from which the surroundings can be observed but at which a watch is not habitually maintained.
37	Timeball	A visual time signal in the form of a ball.
38	Clock	Instrument for measuring time and recording hours.
39	Control	Used to control the flow of traffic within a specified range of an installation.
40	Airship Mooring	Equipment or structure to secure an airship.
41	Stadium	An arena for holding and viewing events.
42	Bus Station	A building where buses and coaches regularly stop to take on and/or let off passengers, especially for long-distance travel.
43	Passenger Terminal Building	A building within a terminal for the loading and unloading of passengers.
44	Sea Rescue Control	A unit responsible for promoting efficient organization of search and rescue services and for coordinating the conduct of search and rescue operations within a search and rescue region.
45	Observatory	A building designed and equipped for making observations of astronomical, meteorological, or other natural phenomena.
46	Ore Crusher	A building or structure used to crush ore.
47	Boathouse	A building or shed, usually built partly over water, for sheltering a boat or boats.
48	Pumping Station	A facility to move solids, liquids or gases

		by means of pressure or suction.
49	Roof Above Navigable Water	A roof that is extending above navigable water, e.g. to protect open cargo holds from rain during loading and unloading. Depending on the vertical clearance vessels can pass under the roof above navigable water.
50	Building Above Navigable Water	The part of a building on land that is extending above navigable water. Depending on the vertical clearance vessels can pass under the building above navigable water.

5.45 Headline

Type: SimpleAttributeType

Definition: Words set at the head of a passage or page to introduce or categorize.

CamelCase: headline

Value type: text

5.46 Height

Type: SimpleAttributeType

Definition: The value of the vertical distance to the highest point of the object, measured from a specified vertical datum.

CamelCase: height

Value type: real

5.47 ID Code

Type: SimpleAttributeType

Definition: Identification code as specified in predefined system. Also called identification number.

CamelCase: iDCode

Value type: text

5.48 Installation Date

Type: SimpleAttributeType

Definition: The date when an item was installed.

CamelCase: installationDate

Value type: date

5.49 Interoperability Identifier

Type: SimpleAttributeType

Definition: A common unique identifier for entities which describe a single real-world feature, and which is used to identify instances of the feature in end-user systems where the feature may be included in multiple data product types.

CamelCase: interoperabilityIdentifier

Value type: URN

5.50 IsSlatted

Type: SimpleAttributeType

Definition: -

CamelCase: isSlatted

Value type: boolean

5.51 Language

Type: SimpleAttributeType

Definition: The method of human communication, either spoken or written, consisting of the use of words in a structured and conventional way.

CamelCase: language

Value type: text

5.52 Light Characteristic

Type: EnumerationType

Definition: The distinct character, such as fixed, flashing, or occulting, which is given to each light to avoid confusion with neighbouring ones.

CamelCase: lightCharacteristic

Value type: enumeration

Listed values:

Code	Label	Definition
1	Fixed	A signal light that shows continuously, in any given direction, with constant luminous intensity and colour.
2	Flashing	A rhythmic light in which the total duration of light in a period is clearly shorter than the total duration of darkness and all the appearances of light are of equal duration.
3	Long-Flashing	A single-flashing light in which a single flash of not less than two seconds duration is regularly repeated.
4	Quick-Flashing	A rhythmic light in which flashes are repeated at a rate of not less than 50 flashes per minutes but less than 80 flashes per minutes. It may be: - Continuous quick-flashing: A quick-flashing light in which a flash is regularly repeated. - Group quick-flashing: A quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.
5	Very Quick-Flashing	A rhythmic light in which flashes are repeated at a rate of not less than 80 flashes per minute but less than 160 flashes per minute. It may be:- Continuous very quick-flashing: A very quick-flashing light in which a flash is regularly repeated.- Group very quick-flashing: A very quick-flashing light in which a group of two or more flashes, which are specified in number, is regularly repeated.
6	Continuous Ultra Quick-Flashing	A rhythmic light in which flashes are regularly repeated at a rate of not less than 160 flashes per minute.
7	Isophased	A light with all durations of light and darkness equal.

8	Occulting	A rhythmic light in which the total duration of light in a period is clearly longer than the total duration of darkness and all the eclipses are of equal duration. It may be: - Single-occulting: An occulting light in which an eclipse is regularly repeated. - Group-occulting: An occulting light in which a group of two or more eclipses, which are specified in number, is regularly repeated. - Composite group-occulting: An occulting light in which a sequence of groups of one or more eclipses, which are specified in number, is regularly repeated, and the groups comprise different numbers of eclipses.
9	Interrupted Quick Flashing	A quick light in which the sequence of flashes is interrupted by regularly repeated eclipses of constant and long duration.
10	Interrupted Very Quick Flashing	A light in which the very rapid alterations of light and darkness are interrupted at regular intervals by eclipses of long duration.
11	Interrupted Ultra Quick-Flashing	A light in which the ultra quick flashes (160 or more per minute) are interrupted at regular intervals by eclipses of long duration.
12	Morse	A rhythmic light in which appearances of light of two clearly different durations are grouped to represent a character or characters in the Morse code.
13	Fixed and Flash	A rhythmic light in which a fixed light is combined with a flashing light of higher luminous intensity.
14	Flash and Long-Flash	A rhythmic light in which a flashing light is combined with a long-flashing light of higher luminous intensity.
15	Occulting and Flash	A rhythmic light in which an occulting light is combined with a flashing light of higher luminous intensity.
16	Fixed and Long-Flash	A rhythmic light in which a fixed light is combined with a long-flashing light of higher luminous intensity.
17	Occulting Alternating	An alternating light in which the total duration of light in each period is clearly longer than the total duration of darkness and in which the intervals of darkness (occultations) are all of equal duration.
18	Long-Flash Alternating	An alternating single-flashing light in which an appearance of light of not less than two seconds duration is regularly repeated.
19	Flash Alternating	An alternating rhythmic light in which the total duration of light in a period is clearly shorter than the total duration of darkness and all the appearances of light are of equal duration.

20	Group Alternating	Occulting light in which the occultations are combined in groups, each group including the same number of occultations, and in which the groups are repeated at regular intervals.
25	Quick-Flash Plus Long-Flash	A rhythmic light in which a group of quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
26	Very Quick-Flash Plus Long-Flash	A rhythmic light in which a group of very quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
27	Ultra Quick-Flash Plus Long-Flash	A rhythmic light in which a group of ultra quick flashes is followed by one or more long flashes in a regularly repeated sequence with a regular periodicity.
28	Alternating	A signal light that shows, in any given direction, two or more colours in a regularly repeated sequence with a regular periodicity.
29	Fixed and Alternating Flashing	A rhythmic light in which a fixed light is combined with a flashing light of higher luminous intensity and different colour.

5.53 Light Visibility

Type: EnumerationType

Definition: The specific visibility of a light, with respect to the light's intensity and ease of recognition.

CamelCase: lightVisibility

Value type: enumeration

Listed values:

Code	Label	Definition
1	High Intensity	Non-marine lights with a higher power than marine lights and visible from well off shore (often 'Aero' lights).
2	Low Intensity	Non-marine lights with lower power than marine lights.
3	Faint	A decrease in the apparent intensity of a light which may occur in the case of partial obstructions.
4	Intensified	A light in a sector is intensified (that is, has longer range than other sectors).
5	Unintensified	A light in a sector is unintensified (that is, has shorter range than other sectors).
6	Visibility Deliberately Restricted	A light sector is deliberately reduced in intensity, for example to reduce its effect on a built-up area.
7	Obscured	Said of the arc of a light sector designated by its limiting bearings in

		which the light is not visible from seaward.
8	Partially Obscured	This value specifies that parts of the sector are obscured.
9	Visible in Line of Range	Lights that must in line to be visible.

5.54 Lighted Aton Change

Type: EnumerationType

Definition: Updates or modifications to light-emitting AtoNs, including changing light characteristics, intensity, or operational status.

CamelCase: lightedAtonChange

Value type: enumeration

Listed values:

Code	Label	Definition
1	Light Unlit	The light is extinguished.
2	Light Unreliable	The light is unreliable due to technical problems.
3	Light Re-Establishment	The re-establishment of a light which was previously announced as either destroyed or temporarily removed.
4	Light Range Reduced	The nominal range of the light is less than the advertised range.
5	Light Without Rhythm	Due to technical problems the light has no more rhythm and is in fixed light mode.
6	Light Out Of Synchronization	The light is no longer synchronized with another light or group of lights.
7	Light Daymark Unreliable	The light daymark is not visible due to damage or fading of colours (out of tolerance with colour recommendations).
8	Light Operating Properly	The light is operating as advertised
9	Sector light Sector obscured	.
10	Front leading/range light Unlit	.
11	Rear leading/range light Unlit	.
12	Front leading/range light Unreliable	.
13	Rear leading/range light Unreliable	.
14	Front leading/range light Light range reduced	.
15	Rear leading/range light Light range reduced	.
16	Front leading/range light without rhythm	.

17	Rear leading/range light without rhythm	.
18	Leading/range lights out of synchronization	.
19	Front leading/range beacon Unreliable	.
20	Rear leading/range beacon Unreliable	.
21	Front leading/range light is operating properly	.
22	Rear leading/range light is operating properly	.
23	Front leading/range beacon restored to normal	.
24	Rear leading/range beacon restored to normal	.

5.55 MMSI Code

Type: SimpleAttributeType

Definition: The Maritime Mobile Service Identity (MMSI) Code is formed of a series of nine digits which are transmitted over the radio path in order to uniquely identify ship stations, ship earth stations, coast stations, coast earth stations, and group calls. These identities are formed in such a way that the identity or part thereof can be used by telephone and telex subscribers connected to the general telecommunications network principally to call ships automatically.

CamelCase: mMSICode

Value type: text

5.56 Major Light

Type: SimpleAttributeType

Definition: A statement expressing if a light is considered to be a major light in terms of ECDIS display in a particular area.

CamelCase: majorLight

Value type: boolean

5.57 Marks Navigational - System Of

Type: EnumerationType

Definition: The system of navigational buoyage a region complies with.

CamelCase: marksNavigationalSystemOf

Value type: enumeration

Listed values:

Code	Label	Definition
1	IALA A	Navigational aids conform to the International Association of Lighthouse Authorities - IALA A system.
2	IALA B	Navigational aids conform to the International Association of Lighthouse Authorities - IALA B system.
9	No System	Navigational aids do not conform to any defined system.

10	Other System	Navigational aids conform to a defined system other than International Association of Lighthouse Authorities - IALA.
11	Main European Inland Waterway Marking System	Navigational aids as required in international, national or regional regulations that contain the same navigational aids as the European Code for Inland Waterways of UNECE, or if there is no regulation for a waterway, navigational aids as recommended in the European Code for Inland Waterways of UNECE
12	Russian Inland Waterway Regulations	Navigational aids conform to the Russian inland waterway regulations.
13	Brazilian National Inland Waterway Regulation	Navigational aids conform to the Brazilian national inland waterway regulation
15	Paraguay-Parana Waterway - Brazilian Complementary Aids	Navigational aids conform to the Brazilian complementary aids on the Paraguay-Parana waterway.

5.58 Maximal Permitted Draught

Type: SimpleAttributeType

Definition: The maximal permitted draught of a vessel or convoy according to the particular article/clause of the applicable law/regulation.

CamelCase: maximalPermittedDraught

Value type: real

5.59 Moire Effect

Type: SimpleAttributeType

Definition: A short range (up to 2km) type of directional light. Sodium lighting gives a yellow background to a screen on which a vertical black line will be seen by an observer on the centre line.

CamelCase: moireEffect

Value type: boolean

5.60 Multiplicity Known

Type: SimpleAttributeType

Definition: The number of features of identical character that exist as a co-located group is or is not known.

CamelCase: multiplicityKnown

Value type: boolean

5.61 Name

Type: SimpleAttributeType

Definition: The individual name of a feature.

CamelCase: name

Value type: text

5.62 Name Usage

Type: EnumerationType

Definition: Classification of the type and display level of the name of a feature in an end-user system.

CamelCase: nameUsage

Value type: enumeration

Listed values:

Code	Label	Definition
1	Default Name Display	The name is intended to be displayed when the end-user system is set to the default name/text display setting.
2	Alternate Name Display	The name is intended to be displayed when the end-user system is set to an alternate name/text display setting, for example an alternate language.
3	No Chart Display	The name or text is not intended to be displayed.

5.63 Nature of Construction

Type: EnumerationType

Definition: The building's primary construction material.

CamelCase: natureOfConstruction

Value type: enumeration

Listed values:

Code	Label	Definition
1	Masonry	Constructed of stones or bricks, usually quarried, shaped, and mortared.
2	Concreted	Constructed of concrete, a material made of sand and gravel that is united by cement into a hardened mass used for roads, foundations, etc.
3	Loose Boulders	Constructed from large stones or blocks of concrete, often placed loosely for protection against waves or water turbulence.
4	Hard Surfaced	Constructed with a surface of hard material, usually a term applied to roads surfaced with asphalt or concrete.
5	Unsurfaced	Constructed with no extra protection, usually a term applied to roads not surfaced with a hard material.
6	Wooden	Constructed from wood.
7	Metal	Constructed from metal.
8	Glass Reinforced Plastic	Constructed from a plastic material strengthened with fibres of glass.
9	Painted	The application of paint to some other construction or natural feature.
10	Framework	Constructed from a lattice framework of,

		often diagonal, intersecting struts.
11	Latticed	A structure of crossed wooden or metal strips usually arranged to form a diagonal pattern of open spaces between the strips.
12	Glass	[1] Any artificial or natural substance having similar properties and composition, as fused borax, obsidian, or the like. [2] Something made of such a substance, as a windowpane.
13	Fiberglass	Constructed from fiberglass.
14	Plastic	Constructed from plastic.

5.64 Number of Features

Type: SimpleAttributeType

Definition: The number of features of identical character that exist as a co-located group.

CamelCase: numberOfFeatures

Value type: integer

5.65 Orientation Uncertainty

Type: SimpleAttributeType

Definition: The best estimate of the accuracy of a bearing.

CamelCase: orientationUncertainty

Value type: real

5.66 Orientation Value

Type: SimpleAttributeType

Definition: The angular distance measured from true north to the major axis of the feature.

CamelCase: orientationValue

Value type: real

5.67 Peak Intensity

Type: SimpleAttributeType

Definition: The maximum luminous intensity of a light during its flash cycle.

CamelCase: peakIntensity

Value type: real

5.68 Pictorial Representation

Type: SimpleAttributeType

Definition: Indicates whether a pictorial representation of the feature is available.

CamelCase: pictorialRepresentation

Value type: text

5.69 Quality of Vertical Measurement

Type: EnumerationType

Definition: The reliability of the value of a sounding.

CamelCase: qualityOfVerticalMeasurement

Value type: enumeration

Listed values:

Code	Label	Definition
1	Depth Known	The depth from the chart datum to the seabed (or to the top of a drying feature) is known.
2	Depth or Least Depth Unknown	The depth from chart datum to the seabed, or the shoalest depth of the feature is unknown.
3	Doubtful Sounding	A depth that may be less than indicated.
4	Unreliable Sounding	A depth that is considered to be an unreliable value.
5	No Bottom Found at Value Shown	Upon investigation the bottom was not found at this depth.
6	Least Depth Known	The shoalest depth over a feature is of known value.
7	Least Depth Unknown, Safe Clearance at Value Shown	The least depth over a feature is unknown, but there is considered to be safe clearance at this depth.
8	Value Reported (Not Surveyed)	Depth value obtained from a report, but not fully surveyed.
9	Value Reported (Not Confirmed)	Depth value obtained from a report, which it has not been possible to confirm.
10	Maintained Depth	The depth at which a channel is kept by human influence, usually by dredging.
11	Not Regularly Maintained	Depths may be altered by human influence, but will not be routinely maintained.

5.70 Radar Band

Type: SimpleAttributeType

Definition: The band code character of the electromagnetic spectrum within which radar wave lengths lie.

CamelCase: radarBand

Value type: text

5.71 Radar Conspicuous

Type: SimpleAttributeType

Definition: A feature which returns a strong radar echo.

CamelCase: radarConspicuous

Value type: boolean

5.72 Scale Minimum

Type: SimpleAttributeType

Definition: The minimum scale at which the feature may be used for example for ECDIS presentation.

CamelCase: scaleMinimum

Value type: integer

5.73 Sector Arc Extension

Type: SimpleAttributeType

Definition: An indication that the default radius of a sector arc is to be extended by 5mm.

CamelCase: sectorArcExtension

Value type: boolean

5.74 Sector Bearing

Type: SimpleAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector bearing specifies the limit of the sector.

CamelCase: sectorBearing

Value type: real

5.75 Sector Line Length

Type: SimpleAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector line length specifies the displayed length of the line, in ground units, defining the limit of the sector.

CamelCase: sectorLineLength

Value type: real

5.76 Signal Duration

Type: SimpleAttributeType

Definition: The time occupied by a single instance of light/sound or eclipse/silence in a signal sequence.

CamelCase: signalDuration

Value type: real

5.77 Signal Generation

Type: EnumerationType

Definition: The mechanism used to generate a fog or light signal.

CamelCase: signalGeneration

Value type: enumeration

Listed values:

Code	Label	Definition
1	Automatically	Signal generation is initiated by a self regulating mechanism such as a timer or light sensor.
2	By Wave Action	The signal is generated by the motion of the sea surface such as a bell in a buoy.
3	By Hand	The signal is generated by a manually operated mechanism such as a hand cranked siren.

4	By Wind	The signal is generated by the motion of air such as a wind driven whistle.
5	Radio Activated	Activated by radio signal.
6	Call Activated	Activated by making a call to a manned station.

5.78 Signal Group

Type: SimpleAttributeType

Definition: The number of signals, the combination of signals or the morse character(s) within one period of full sequence.

CamelCase: signalGroup

Value type: text

5.79 Signal Period

Type: SimpleAttributeType

Definition: The time occupied by an entire cycle of intervals of light and eclipse.

CamelCase: signalPeriod

Value type: real

5.80 Signal Status

Type: EnumerationType

Definition: The indication of an element of a signal sequence being a period of light/sound or eclipse/silence.

CamelCase: signalStatus

Value type: enumeration

Listed values:

Code	Label	Definition
1	Lit/Sound	The indication of an element of a signal sequence being a period of light or sound.
2	Eclipsed/Silent	The indication of an element of a signal sequence being a period of eclipse or silence.

5.81 Source

Type: SimpleAttributeType

Definition: The publication, document, or reference work from which information comes or is acquired.

CamelCase: source

Value type: text

5.82 Source Date

Type: SimpleAttributeType

Definition: The production date of the source; for example the date of measurement.

CamelCase: sourceDate

Value type: date

5.83 Status

Type: EnumerationType

Definition: The condition of an object at a given instant in time.

CamelCase: status

Value type: enumeration

Listed values:

Code	Label	Definition
1	Permanent	Intended to last or function indefinitely.
2	Occasional	Acting on special occasions; happening irregularly.
3	Recommended	Presented as worthy of confidence, acceptance, use, etc.
4	Not in Use	Use has ceased, but the facility still exists intact; disused.
5	Periodic/Intermittent	Recurring at intervals.
6	Reserved	Set apart for some specific use.
7	Temporary	Meant to last only for a time.
8	Private	Administered by an individual or corporation, rather than a State or a public body.
9	Mandatory	Compulsory; enforced.
11	Extinguished	No longer lit.
12	Illuminated	Lit by floodlights, strip lights, etc.
13	Historic	Famous in history; of historical interest.
14	Public	Belonging to, available to, used or shared by, the community as a whole and not restricted to private use.
15	Synchronized	Occur at a time, coincide in point of time, be contemporary or simultaneous.
16	Watched	Looked at or observed over a period of time especially so as to be aware of any movement or change.
17	Unwatched	Usually automatic in operation, without any permanently-stationed personnel to superintend it.
18	Existence Doubtful	A feature that has been reported but has not been definitely determined to exist.
19	On Request	When you ask for it.
20	Drop Away	To become lower in level.
21	Rising	To become higher in level.
22	Increasing	Becoming larger in magnitude.

23	Decreasing	Becoming smaller in magnitude.
24	Strong	Not easily broken or destroyed.
25	Good	In a satisfactory condition to use.
26	Moderately	Fairly but not very.
27	Poor	Not as good as it could be or should.
28	Buoyed	Marked by buoys.
29	Fully Operational	Entire observation platform is operating in accordance with, or exceeding, manufacturer specifications.
30	Partially Operational	At least one instrument that is part of an observation platform is not operating to manufacturer specification.
31	Drifting	Floating platform at the mercy of environmental elements, whether intentional or not.
32	Broken	Fractured or in pieces.
33	Offline	Observation platform is intentionally not reporting an environmental observation.
34	Discontinued	Observation station, suite of instruments, or an individual instrument, for a particular location, has been removed and is no longer at the particular location.
35	Manual Observation	Observations made by a human observer.
36	Unknown Status	Status of an observation platform, suite of instruments, or individual instrument is not known or unspecified.
37	Confirmed	Made certain as to truth, accuracy, validity, availability, etc.
38	Candidate	Item selected for an action.
39	Under Modification	Item that is in the process of being modified.
41	Under Removal / Deletion	Item in the process of being removed or deleted.
42	Removed / Deleted	Item that has been removed or deleted.
43	Candidate for Modification	Item selected for modification.

5.84 Technique of Vertical Measurement

Type: EnumerationType

Definition: Survey method used to obtain depth information.

CamelCase: techniqueOfVerticalMeasurement

Value type: enumeration

Listed values:

Code	Label	Definition
1	Found by Echo Sounder	The depth was determined by using an instrument that determines depth of water by measuring the time interval between emission of a sonic or ultrasonic signal and return of its echo from the bottom.
2	Found by Side Scan Sonar	The depth was computed from a record produced by active sonar in which fixed acoustic beams are directed into the water perpendicularly to the direction of travel to scan the seabed and generate a record of the seabed configuration.
3	Found by Multi Beam	The depth was determined by using a wide swath echo sounder that uses multiple beams to measure depths directly below and transverse to the ship's track.
4	Found by Diver	The depth was determined by a person skilled in the practice of diving.
5	Found by Lead Line	The depth was determined by using a line, graduated with attached marks and fastened to a sounding lead.
6	Swept by Wire-Drag	The given area was determined to be free from navigational dangers to a certain depth by towing a buoyed wire at the desired depth by two launches, or a least depth was identified using the same technique.
7	Found by Laser	The depth was determined by using an instrument that measures distance by emitting timed pulses of laser light and measuring the time between emission and reception of the reflected pulses.
8	Swept by Vertical Acoustic System	The given area has been swept using a system comprised of multiple echo sounder transducers attached to booms deployed from the survey vessel.
9	Found by Electromagnetic Sensor	The depth was determined by using an instrument that compares electromagnetic signals.
10	Photogrammetry	The science or art of obtaining reliable measurements from photographs.
11	Satellite Imagery	The depth was determined by using instruments placed aboard an artificial satellite.
12	Found by Levelling	The depth was determined by using levelling techniques to find the elevation of the point relative to a datum.
13	Swept by Side Scan Sonar	The given area was determined to be free from navigational dangers to a certain depth by towing a side scan sonar.

14	Computer Generated	The sounding was determined from a bottom model constructed using a computer.
15	Found by LIDAR	The depth was measured by using an instrument that measures distance by emitting timed pulses of laser light and measuring the time between emission and reception of the reflected pulses.
16	Synthetic Aperture Radar	A radar with a synthetic aperture antenna which is composed of a large number of elementary transducing elements. The signals are electronically combined into a resulting signal equivalent to that of a single antenna of a given aperture in a given direction.
17	Hyperspectral Imagery	Term used to describe the imagery derived from subdividing the electromagnetic spectrum into very narrow bandwidths. These narrow bandwidths may be combined with or subtracted from each other in various ways to form images useful in precise terrain or target analysis.
18	Mechanically Swept	The given area was determined to be free from navigational dangers to a certain depth by towing a line or object below the surface at the desired depth; or least depth(s) and position(s) within an area was identified using the same technique.

5.85 Text

Type: SimpleAttributeType

Definition: A non-formatted digital text string.

CamelCase: text

Value type: text

5.86 Time of Day End

Type: SimpleAttributeType

Definition: The time corresponding to the end of an active period.

CamelCase: timeOfDayEnd

Value type: time

5.87 Time of Day Start

Type: SimpleAttributeType

Definition: The time corresponding to the start of an active period.

CamelCase: timeOfDayStart

Value type: time

5.88 Topmark/Daymark Shape

Type: EnumerationType

Definition: The shape a topmark or daymark exhibits.

CamelCase: topmarkDaymarkShape

Value type: enumeration

Listed values:

Code	Label	Definition
1	Cone (Point Up)	Is where the vertex points up. A cone is a solid figure generated by straight lines drawn from a fixed point (the vertex) to a circle in a plane not containing the vertex. Cones are commonly used as International Association of Lighthouse Authorities - IALA topmarks, lateral.
2	Cone (Point Down)	Is where the vertex points down. A cone is a solid figure generated by straight lines drawn from a fixed point (the vertex) to a circle in a plane not containing the vertex. Cones are commonly used as International Association of Lighthouse Authorities - IALA topmarks, lateral.
3	Sphere	A curved surface all points of which are equidistant from a fixed point within, called the centre.
4	2 Spheres	Two spheres, one above the other. Two black spheres are commonly used as an International Association of Lighthouse Authorities - IALA topmark (isolated danger).
5	Cylinder	A solid geometrical figure generated by straight lines fixed in direction and describing with one of point a closed curve, especially a circle (in which case the figure is circular cylinder, it's ends being parallel circles). Cylinders are commonly used as International Association of Lighthouse Authorities - IALA topmarks lateral.
6	Board	Usually of rectangular shape, made from timber or metal and used to provide a contrast with the natural background of a daymark. The actual daymark is often painted on to this board.
7	X-Shaped	Having a shape or a cross-section like the capital letter X. An x-shape as an International Association of Lighthouse Authorities - IALA topmark should be 3 dimensional in shape. It is made of at least three crossed bars.
8	Upright Cross	A cross with one vertical member and one horizontal member; that is, similar in shape to the character '+'.
9	Cube (Point Up)	A cube standing on one of its vertexes. A cube is a solid contained by six equal squares, a regular hexahedron.
10	2 Cones (Point to Point)	2 cones, one above the other, with their vertices together in the centre.
11	2 Cones (Base to Base)	2 cones, one above the other, with their bases together in the centre and their vertices pointing up and down.

12	Rhombus	A plane figure having four equal sides and equal opposite angles (two acute and two obtuse); an oblique equilateral parallelogram.
13	2 Cones (Points Upward)	2 cones, one above the other, with their vertices pointing up.
14	2 Cones (Points Downward)	2 cones, one above the other, with their vertices pointing down.
15	Besom (Point Up)	Besom: A bundle of rods or twigs. Perch: A staff placed on top of a buoy, rock or shoal as a mark for navigation. A besom, point up is where the thicker (untied) end of the besom is at the bottom.
16	Besom (Point Down)	Besom: A bundle of rods or twigs. Perch: A staff placed on top of a buoy, rock or shoal as a mark for navigation. A besom, point down is where the thinner (tied) end of the besom is at the bottom.
17	Flag	A flag mounted on a short pole.
18	Sphere Over a Rhombus	A sphere located above a rhombus.
19	Square	A plane figure with four right angles and four equal straight sides.
20	Rectangle (Horizontal)	Where the two longer opposite sides are standing horizontally. A rectangle is a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length.
21	Rectangle (Vertical)	Where the two longer opposite sides are standing vertically. A rectangle is a plane figure with four right angles and four straight sides, opposite sides being parallel and equal in length.
22	Trapezium (Up)	A quadrilateral having one pair of opposite sides parallel, and which stands on its longer parallel side.
23	Trapezium (Down)	A quadrilateral having one pair of opposite sides parallel, and which stands on its shorter parallel side.
24	Triangle (Point Up)	A figure having three angles and three sides, and which has a vertex at the top.
25	Triangle (Point Down)	A figure having three angles and three sides, and which has a side at the top.
26	Circle	A perfectly round plane figure whose circumference is everywhere equidistant from its centre.
27	Two Upright Crosses (One Over the Other)	Two upright crosses, generally vertically disposed one above the other.
28	T-Shape	Having a shape like the capital letter T.
29	Triangle Pointing Up Over a Circle	A triangle, vertex uppermost, located above a circle.

30	Upright Cross Over a Circle	An upright cross located above a circle.
31	Rhombus Over a Circle	A rhombus located above a circle.
32	Circle Over a Triangle Pointing Up	A circle located over a triangle, vertex uppermost.
33	Other Shape (See Shape Information)	An uncommon and/or non-standardized shape as textually described using an associated attribute.
34	Tubular	Having the form of or consisting of a tube.

5.89 Traffic Flow

Type: EnumerationType

Definition: Direction of vessels passing a reference point.

CamelCase: trafficFlow

Value type: enumeration

Listed values:

Code	Label	Definition
1	Inbound	Traffic flow in a general direction toward a port or similar destination.
2	Outbound	Traffic flow in a general direction away from a port or similar point of origin.
3	One-Way	Traffic flow in one general direction only.
4	Two-Way	Traffic flow in two generally opposite directions.

5.90 Type of Buoy

Type: SimpleAttributeType

Definition: Type equipment used as a buoy in a particular installation.

CamelCase: typeOfBuoy

Value type: text

5.91 Uncertainty Fixed

Type: SimpleAttributeType

Definition: The best estimate of the fixed horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances.

CamelCase: uncertaintyFixed

Value type: real

5.92 Uncertainty Variable Factor

Type: SimpleAttributeType

Definition: The factor to be applied to the variable component of an uncertainty equation so as to provide the best estimate of the variable horizontal or vertical accuracy component for positions, depths, heights, vertical distances and vertical clearances.

CamelCase: uncertaintyVariableFactor

Value type: real

5.93 Value of Nominal Range

Type: SimpleAttributeType

Definition: The luminous range of a light in a homogenous atmosphere in which the meteorological visibility is 10 sea miles.

CamelCase: valueOfNominalRange

Value type: real

5.94 Vertical Accuracy

Type: SimpleAttributeType

Definition: The degree to which a vertical measurement is accurate, typically referring to the accuracy of an object's position in the vertical plane (height or depth)

CamelCase: verticalAccuracy

Value type: real

5.95 Vertical Datum

Type: EnumerationType

Definition: The reference level used for expressing the vertical measurements of points on the earth's surface. Also called datum level, reference plane, levelling datum, datum for sounding reduction, datum for heights.

CamelCase: verticalDatum

Value type: enumeration

Listed values:

Code	Label	Definition
1	Mean Low Water Springs	The average height of the low waters of spring tides. This level is used as a tidal datum in some areas. Also called spring low water.
2	Mean Lower Low Water Springs	The average height of lower low water springs at a place.
3	Mean Sea Level	The average height of the surface of the sea at a tide station for all stages of the tide over a 19-year period, usually determined from hourly height readings measured from a fixed predetermined reference level.
4	Lowest Low Water	An arbitrary level conforming to the lowest tide observed at a place, or some what lower.
5	Mean Low Water	The average height of all low waters at a place over a 19-year period.
6	Lowest Low Water Springs	An arbitrary level conforming to the lowest water level observed at a place at spring tides during a period of time shorter than 19 years.
7	Approximate Mean Low Water Springs	An arbitrary level, usually within 0.3m from that of Mean Low Water Springs (MLWS).
8	Indian Spring Low Water	An arbitrary tidal datum approximating the level of the mean of the lower low

		water at spring tides. It was first used in waters surrounding India.
9	Low Water Springs	An arbitrary level, approximating that of mean low water springs (MLWS).
10	Approximate Lowest Astronomical Tide	An arbitrary level, usually within 0.3m from that of Lowest Astronomical Tide (LAT).
11	Nearly Lowest Low Water	An arbitrary level approximating the lowest water level observed at a place, usually equivalent to the Indian Spring Low Water (ISLW).
12	Mean Lower Low Water	The average height of the lower low waters at a place over a 19-year period.
13	Low Water	The lowest level reached at a place by the water surface in one oscillation. Also called low tide.
14	Approximate Mean Low Water	An arbitrary level, usually within 0.3m from that of Mean Low Water (MLW).
15	Approximate Mean Lower Low Water	An arbitrary level, usually within 0.3m from that of Mean Lower Low Water (MLLW).
16	Mean High Water	The average height of all high waters at a place over a 19-year period.
17	Mean High Water Springs	The average height of the high waters of spring tides. Also called spring high water.
18	High Water	The highest level reached at a place by the water surface in one oscillation.
19	Approximate Mean Sea Level	An arbitrary level, usually within 0.3m from that of Mean Sea Level (MSL).
20	High Water Springs	An arbitrary level, approximating that of mean high water springs (MHWS).
21	Mean Higher High Water	The average height of higher high waters at a place over a 19-year period.
22	Equinoctial Spring Low Water	The level of low water springs near the time of an equinox.
23	Lowest Astronomical Tide	The lowest tide level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions.
24	Local Datum	An arbitrary datum defined by a local harbour authority, from which levels and tidal heights are measured by this authority.
25	International Great Lakes Datum 1985	A vertical reference system with its zero based on the mean water level at Rimouski/Pointe-au-Pere, Quebec, over the period 1970 to 1988.
26	Mean Water Level	The average of all hourly water levels

		over the available period of record.
27	Lower Low Water Large Tide	The average of the lowest low waters, one from each of 19 years of observations.
28	Higher High Water Large Tide	The average of the highest high waters, one from each of 19 years of observations.
29	Nearly Highest High Water	An arbitrary level approximating the highest water level observed at a place, usually equivalent to the high water springs.
30	Highest Astronomical Tide	The highest tidal level which can be predicted to occur under average meteorological conditions and under any combination of astronomical conditions.
31	Local Low Water Reference Level	Low water reference level of the local area.
32	Local High Water Reference Level	High water reference level of the local area.
33	Local Mean Water Reference Level	Mean water reference level of the local area.
34	Equivalent Height of Water (German GIW)	A low water level which is the result of a defined low water discharge - called equivalent discharge.
35	Highest Shipping Height of Water (German HSW)	Upper limit of water levels where navigation is allowed.
36	Reference Low Water Level According to Danube Commission	The water level at a discharge, which is exceeded 94 % of the year within a period of 30 years.
37	Highest Shipping Height of Water According to Danube Commission	The water level at a discharge, which is exceeded 1% of the year within a period of 30 years.
38	Dutch River Low Water Reference Level (OLR)	The water level at a discharge, which is exceeded 95% of the year within a period of 20 years.
39	Russian Project Water Level	Conditional low water level with established probability.
40	Russian Normal Backwater Level	Highest water level derived from the upper backwater stream in watercourse or reservoir under the normal operational conditions.
41	Ohio River Datum	The Ohio River datum.
43	Dutch High Water Reference Level	Dutch High Water Reference Level.
44	Baltic Sea Chart Datum 2000	The datum refers to each Baltic country's realization of the European Vertical Reference System (EVRS) with land-uplift epoch 2000, which is connected to the Normaal Amsterdams Peil (NAP).
45	Dutch Estuary Low Water Reference	Dutch Estuary Low Water Reference

	Level (OLW)	Level (OLW)
46	International Great Lakes Datum 2020	The 2020 update to the International Great Lakes Datum, the official reference system used to measure water level heights in the Great Lakes, connecting channels, and the St. Lawrence River system.
47	Sea Floor	The bottom of the ocean and seas where there is a generally smooth gentle gradient. Also referred to as sea bed (sometimes seabed or sea-bed), and sea bottom.
48	Sea Surface	A two-dimensional (in the horizontal plane) field representing the air-sea interface, with high-frequency fluctuations such as wind waves and swell, but not astronomical tides, filtered out.
49	Hydrographic Zero	A vertical reference near the lowest astronomical tide (LAT), below which the sea level falls only very exceptionally.

5.96 Vertical Length

Type: SimpleAttributeType

Definition: The total vertical length of a feature.

CamelCase: verticalLength

Value type: real

5.97 Virtual AIS Aid to Navigation Type

Type: EnumerationType

Definition: A purpose of a virtual AIS Aid to Navigation.

CamelCase: virtualAISaidToNavigationType

Value type: enumeration

Listed values:

Code	Label	Definition
1	North Cardinal	Indicates that it should be passed to the north side of the aid.
2	East Cardinal	Indicates that it should be passed to the east side of the aid.
3	South Cardinal	Indicates that it should be passed to the south side of the aid.
4	West Cardinal	Indicates that it should be passed to the west side of the aid.
5	Port Lateral	Indicates the port boundary of a navigational channel or suggested route when proceeding in the conventional direction of buoyage.
6	Starboard Lateral	Indicates the starboard boundary of a navigational channel or suggested route

		when proceeding in the conventional direction of buoyage.
7	Preferred Channel to Port	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified port-hand lateral mark.
8	Preferred Channel to Starboard	At a point where a channel divides, when proceeding in the conventional direction of buoyage, the preferred channel (or primary route) is indicated by a modified starboard-hand lateral mark.
9	Isolated Danger	A mark used alone to indicate a dangerous reef or shoal. The mark may be passed on either hand.
10	Safe Water	Indicates that there is navigable water around the mark.
11	Special Purpose	A special purpose aid is primarily used to indicate an area or feature, the nature of which is apparent from reference to a chart, Sailing Directions or Notice to Mariners
12	New Danger Marking	A mark used to indicate the existence of a recently identified new danger, such as a wreck.

5.98 Visual Prominence

Type: EnumerationType

Definition: The extent to which a feature, either natural or artificial, is visible from seaward.

CamelCase: visualProminence

Value type: enumeration

Listed values:

Code	Label	Definition
1	Visually Conspicuous	Term applied to a feature either natural or artificial which is distinctly and notably visible from seaward.
2	Not Visually Conspicuous	An object that may be visible from seaward, but cannot be used as a fixing mark and is not conspicuous.
3	Prominent	Objects which are easily identifiable, but do not justify being classed as conspicuous.

5.99 Wave Length Value

Type: SimpleAttributeType

Definition: The distance between two successive peaks (or other points of identical phase) on an electromagnetic wave.

CamelCase: waveLengthValue

Value type: real

6 Complex attributes

6.1 Change Details

Type: ComplexAttributeType

Definition: -

CamelCase: ChangeDetails

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Aton Commissioning		1. Buoy Establishment 2. Light Establishment 3. Beacon Establishment 4. Audible Signal Establishment 5. Fog Signal Establishment 6. AIS Transmitter Establishment 7. V-AIS Establishment 8. RACON Establishment 9. DGPS Station Establishment 10. ELORAN Station Establishment 11. DGLONASS Station Establishment 12. E-Chayka Station Establishment 13. EGNOS establishment	EN	0, 1
Aton Removal		1. Buoy Removal 2. Buoy Temporary Removal 3. Light Removal 4. Light Temporary Removal 5. Beacon Removal 6. Beacon Temporary Removal 7. Fog Signal Removal 8. Fog Signal Temporary Removal 9. Audible Signal Removal 10. Audible Signal Temporary Removal	EN	0, 1

		11. V-AIS Removal 12. V-AIS Temporary Removal 13. RACON signal removal 14. RACON Temporary Removal 15. DGPS removal 16. DGPS temporary removal 17. EGNOS removal 18. EGNOS temporary removal 19. LORAN C Station Removal 20. LORAN C Station Temporary Removal 21. eLORAN removal 22. eLORAN temporary removal 23. Chayka Station Removal 24. Chayka Station Temporary Removal 25. E-Chayka Station Removal 26. E-Chayka Station Temporary Removal		
Aton Replacement		1. Buoy Change 2. Buoy Temporary Change 3. Light Change 4. Light Temporary Change 5. Sector Light Change 6. Sector Light Temporary Change 7. Beacon Change 8. Beacon Temporary Change 9. Fog Signal Change 10. Fog Signal Temporary Change 11. Audible Signal Change 12. Audible Signal Temporary Change	EN	0, 1

		13. V-AIS Change 14. V-AIS Temporary Change 15. RACON signal change 16. RACON Temporary Change		
Fixed Aton Change		1. Beacon Missing 2. Beacon Damaged 3. Light beacon Unlit 4. Light beacon Unreliable 5. Light beacon Not synchronized 6. Light beacon damaged 7. Beacon Topmark Missing 8. Beacon Topmark Damaged 9. Beacon Daymark Unreliable 10. Floodlit beacon Unlit 11. Beacon Restored To Normal	EN	0, 1
Floating Aton Change		1. Buoy Adrift 2. Buoy Damaged 3. Buoy Daymark Unreliable 4. Buoy Destroyed 5. Buoy Missing 6. Buoy Move 7. Buoy off Position 8. Buoy re-establishment 9. Buoy Restored to Normal 10. Buoy Topmark Damaged 11. Buoy Topmark Missing 12. Buoy Will Be Withdrawn 13. Buoy Withdrawn 14. Decommissioned for winter 15. Lifted for Winter	EN	0, 1

		16. Light buoy Light damaged 17. Light buoy Light not synchronized 18. Light buoy Light unlit 19. Light buoy Light unreliable 20. Marine Aids to Navigation Unreliable 21. Recommissioned for navigation season 22. Replaced by Winter Spar 23. Seasonal Decommissioning Complete 24. Seasonal Decommissioning in Progress 25. Seasonal recommissioning complete 26. Seasonal recommissioning in progress		
Audible Signal Aton Change		1. Audible Signal Out Of Service 2. Fog Signal Out Of Service 3. Audible Signal Operating Properly 4. Fog Signal Operating Properly	EN	0, 1
Lighted Aton Change		1. Light Unlit 2. Light Unreliable 3. Light Re-Establishment 4. Light Range Reduced 5. Light Without Rhythm 6. Light Out Of Synchronization 7. Light Daymark Unreliable 8. Light Operating Properly 9. Sector light Sector obscured 10. Front leading/range light Unlit 11. Rear leading/range light Unlit 12. Front leading/range light	EN	0, 1

		Unreliable 13. Rear leading/range light Unreliable 14. Front leading/range light Light range reduced 15. Rear leading/range light Light range reduced 16. Front leading/range light without rhythm 17. Rear leading/range light without rhythm 18. Leading/range lights out of synchronization 19. Front leading/range beacon Unreliable 20. Rear leading/range beacon Unreliable 21. Front leading/range light is operating properly 22. Rear leading/range light is operating properly 23. Front leading/range beacon restored to normal 24. Rear leading/range beacon restored to normal		
Electronic Aton Change		1. AIS Transmitter Out Of Service 2. AIS Transmitter Unreliable 3. AIS Transmitter Operating Properly 4. V-AIS Out Of Service 5. V-AIS Unreliable 6. V-AIS Operating Properly 7. RACON Out Of Service 8. RACON Unreliable 9. RACON Operating Properly 10. DGPS Out Of Service 11. DGPS Operating Properly 12. DGPS Unreliable 13. LORAN C operating properly	EN	0, 1

		14. LORAN C unreliable		
		15. LORAN C out of service		
		16. ELORAN Operating Properly		
		17. ELORAN Unreliable		
		18. ELORAN Out Of Service		
		19. DGLOANSS operating properly		
		20. DGLOANSS unreliable		
		21. DGLOANSS out of service		
		22. Chayka Operating Properly		
		23. Chayka Unreliable		
		24. Chayka Out Of Service		
		25. E-Chayka Operating Properly		
		26. E-Chayka Unreliable		
		27. E-Chayka Out Of Service		
		28. EGNOS Operating Properly		
		29. EGNOS Unreliable		
		30. EGNOS Out Of Service		

6.2 Obscured Sector

Type: ComplexAttributeType

Definition: -

CamelCase: ObscuredSector

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Sector Limit			C	1, 1
Sector Limit One			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1
Sector Limit Two			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1

Sector Information			C	0, 1
Language			(S) TE	0, 1
Text			(S) TE	1, 1

6.3 Directional Character

Type: ComplexAttributeType

Definition: A directional light is a light illuminating a sector of very narrow angle and intended to mark a direction to follow.

CamelCase: directionalCharacter

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Moire Effect			BO	0, 1
Orientation			C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1

6.4 Feature Name

Type: ComplexAttributeType

Definition: Provides the name of an entity, defines the national language of the name, and provides the option to display the name at various system display settings.

CamelCase: featureName

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Language			TE	1, 1
Name			TE	1, 1
Name Usage		1. Default Name Display 2. Alternate Name Display 3. No Chart Display	EN	0, 1

6.5 Fixed Date Range

Type: ComplexAttributeType

Definition: An active period of a single fixed event or occurrence, as the date range between discrete start and end dates.

CamelCase: fixedDateRange

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
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Date End			TD	0, 1
Date Start			TD	0, 1
Time of Day End			TI	0, 1
Time of Day Start			TI	0, 1

6.6 Horizontal Position Uncertainty

Type: ComplexAttributeType

Definition: The best estimate of the accuracy of a position.

CamelCase: horizontalPositionUncertainty

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Uncertainty Fixed			RE	1, 1
Uncertainty Variable Factor			RE	0, 1

6.7 Information

Type: ComplexAttributeType

Definition: Textual information about the feature. The information may be provided as a string of text or as a file name of a single external text file that contains the text.

CamelCase: information

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
File Locator			TE	0, 1
File Reference			TE	0, 1
Headline			TE	0, 1
Language			TE	1, 1
Text			TE	0, 1

6.8 Light Sector

Type: ComplexAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference.

CamelCase: lightSector

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Colour		1. White	EN	1, 99

		2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown 9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink		
Directional Character			C	0, 1
Moire Effect			(S) BO	0, 1
Orientation			(S) C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1
Light Visibility		1. High Intensity 2. Low Intensity 3. Faint 4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range	EN	0, 99
Sector Limit			C	0, 1
Sector Limit One			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1

Sector Limit Two			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1
Value of Nominal Range			RE	0, 1
Sector Information			C	0, 99
Language			(S) TE	0, 1
Text			(S) TE	1, 1
Sector Arc Extension			BO	0, 1

6.9 Multiplicity of Features

Type: ComplexAttributeType

Definition: The number of features of identical character that exist as a colocated group.

CamelCase: multiplicityOfFeatures

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Multiplicity Known			BO	1, 1
Number of Features			IN	0, 1

6.10 Orientation

Type: ComplexAttributeType

Definition: The angular distance measured from true north to the major axis of the feature.

CamelCase: orientation

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Orientation Uncertainty			RE	0, 1
Orientation Value			RE	1, 1

6.11 Periodic Date Range

Type: ComplexAttributeType

Definition: The active period of a recurring event or occurrence.

CamelCase: periodicDateRange

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Date End			TD	1, 1

Date Start			TD	1, 1
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6.12 Radar Wave Length

Type: ComplexAttributeType

Definition: The distance between two successive peaks (or other points of identical phase) on an electromagnetic wave in the radar band of the electromagnetic spectrum.

CamelCase: radarWaveLength

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Radar Band			TE	1, 1
Wave Length Value			RE	1, 1

6.13 Rhythm of Light

Type: ComplexAttributeType

Definition: The sequence of times occupied by intervals of light/sound and eclipse/silence for all light characteristics or sound signals.

CamelCase: rhythmOfLight

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-Flashing 7. Isophased 8. Occulting 9. Interrupted Quick Flashing 10. Interrupted Very Quick Flashing 11. Interrupted Ultra Quick-Flashing 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash	EN	1, 1

		15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing		
Signal Group			TE	0, 10
Signal Period			RE	0, 1
Signal Sequence			C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1

6.14 Sector Characteristics

Type: ComplexAttributeType

Definition: Describes the characteristics of a light sector.

CamelCase: sectorCharacteristics

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Light Characteristic		1. Fixed 2. Flashing 3. Long-Flashing 4. Quick-Flashing 5. Very Quick-Flashing 6. Continuous Ultra Quick-	EN	1, 1

		Flashing 7. Isophased 8. Occulting 9. Interrupted Quick Flashing 10. Interrupted Very Quick Flashing 11. Interrupted Ultra Quick-Flashing 12. Morse 13. Fixed and Flash 14. Flash and Long-Flash 15. Occulting and Flash 16. Fixed and Long-Flash 17. Occulting Alternating 18. Long-Flash Alternating 19. Flash Alternating 20. Group Alternating 25. Quick-Flash Plus Long-Flash 26. Very Quick-Flash Plus Long-Flash 27. Ultra Quick-Flash Plus Long-Flash 28. Alternating 29. Fixed and Alternating Flashing		
Light Sector			C	1, 10
Colour		1. White 2. Black 3. Red 4. Green 5. Blue 6. Yellow 7. Grey 8. Brown	(S) EN	1, 99

		9. Amber 10. Violet 11. Orange 12. Magenta 13. Pink		
Directional Character			(S) C	0, 1
Moire Effect			(S) BO	0, 1
Orientation			(S) C	1, 1
Orientation Uncertainty			(S) RE	0, 1
Orientation Value			(S) RE	1, 1
Light Visibility		1. High Intensity 2. Low Intensity 3. Faint 4. Intensified 5. Unintensified 6. Visibility Deliberately Restricted 7. Obscured 8. Partially Obscured 9. Visible in Line of Range	(S) EN	0, 99
Sector Limit			(S) C	0, 1
Sector Limit One			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1
Sector Limit Two			(S) C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1
Value of Nominal Range			(S) RE	0, 1
Sector Information			(S) C	0, 99
Language			(S) TE	0, 1
Text			(S) TE	1, 1

Sector Arc Extension			(S) BO	0, 1
Signal Group			TE	0, 10
Signal Period			RE	0, 1
Signal Sequence			C	0, 10
Signal Duration			(S) RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	(S) EN	1, 1

6.15 Sector Information

Type: ComplexAttributeType

Definition: Additional textual information about a light sector.

CamelCase: sectorInformation

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Language			TE	0, 1
Text			TE	1, 1

6.16 Sector Limit

Type: ComplexAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. The sector limit specifies the limits of the sector In a clockwise direction around the central feature (for example a light).

CamelCase: sectorLimit

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Sector Limit One			C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1
Sector Limit Two			C	1, 1
Sector Bearing			(S) RE	1, 1
Sector Line Length			(S) RE	0, 1

6.17 Sector Limit One

Type: ComplexAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector limit one specifies the first limit of the sector. The order of sector limit one and sector limit two is clockwise around the central feature (for

example a light).

CamelCase: sectorLimitOne

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Sector Bearing			RE	1, 1
Sector Line Length			RE	0, 1

6.18 Sector Limit Two

Type: ComplexAttributeType

Definition: A sector is the part of a circle between two straight lines drawn from the centre to the circumference. Sector limit two specifies the second limit of the sector. The order of sector limit one and sector limit two is clockwise around the central feature (for example a light).

CamelCase: sectorLimitTwo

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Sector Bearing			RE	1, 1
Sector Line Length			RE	0, 1

6.19 Signal Sequence

Type: ComplexAttributeType

Definition: The sequence of times occupied by intervals of light and eclipse for all light characteristics.

CamelCase: signalSequence

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Signal Duration			RE	1, 1
Signal Status		1. Lit/Sound 2. Eclipsed/Silent	EN	1, 1

6.20 Vertical Uncertainty

Type: ComplexAttributeType

Definition: The best estimate of the vertical accuracy of depths, heights, vertical distances and vertical clearances.

CamelCase: verticalUncertainty

Attribute bindings:

S-10x Attribute	S-57 Acronym.	Allowable Encoding Value	Type	Multiplicity
Uncertainty Fixed			RE	1, 1

Uncertainty Variable Factor			RE	0, 1
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7 Roles

7.1 Parent

Type: Role

Definition: -

CamelCase: parentFihoXyz

7.2 Child

Type: Role

Definition: -

CamelCase: child

7.3 Virtual AIS Broadcast By

Type: Role

Definition: -

CamelCase: virtualAISbroadcastBy

7.4 Synthetic AIS Broadcast By

Type: Role

Definition: -

CamelCase: syntheticAISbroadcastBy

7.5 Physical AIS Broadcast By

Type: Role

Definition: -

CamelCase: physicalAISbroadcastBy

7.6 Peer Aton Aggregation

Type: Role

Definition: -

CamelCase: peerAtonAggregation

7.7 Peer Aton Association

Type: Role

Definition: -

CamelCase: peerAtonAssociation

7.8 Navigable Track

Type: Role

Definition: The role given to the navigable part of the navigation line.

CamelCase: navigableTrack

7.9 Danger

Type: Role

Definition: A signal or message warning of the presence of a danger to navigation.

CamelCase: danger

7.10 Topmark Part

Type: Role

Definition: -

CamelCase: topmarkPart

7.11 Status Part

Type: Role

Definition: -

CamelCase: Statuspart

7.12 Virtual AIS Broadcasts

Type: Role

Definition: -

CamelCase: virtualAISbroadcasts

7.13 Synthetic AIS Broadcasts

Type: Role

Definition: -

CamelCase: syntheticAISbroadcasts

7.14 Physical AIS Broadcasts

Type: Role

Definition: -

CamelCase: physicalAISbroadcasts

7.15 Aton Aggregation By

Type: Role

Definition: -

CamelCase: atonAggregationBy

7.16 Aton Association By

Type: Role

Definition: -

CamelCase: atonAssociationBy

7.17 Navigation Line

Type: Role

Definition: The role given to the navigation line(s) that is generally formed between two or more objects, or by one object and a bearing.

CamelCase: navigationLineFihoXyz

7.18 Marking Aton

Type: Role

Definition: -

CamelCase: markingAton

7.19 Buoy Part

Type: Role

Definition: -

CamelCase: buoyPart

8 InformationAssociations

8.1 Aton Status

Type: InformationAssociationType

Definition:

CamelCase: Atonstatus

Association	Statuspart	AtonStatusInformation	0..1	<i>AidsToNavigation</i>
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9 FeatureAssociations

9.1 Buoy Topmark

Type: FeatureAssociationType

Definition:

CamelCase: BuoyTopmark

References:

Association	topmarkPart	Topmark	0..*	<i>GenericBuoy</i>
Association	buoyPart	GenericBuoy	1..1	<i>Topmark</i>

9.2 Structure/Equipment

Type: FeatureAssociationType

Definition: A feature association for the binding between a navigation aid equipment feature and the structure that supports it.

CamelCase: StructureEquipment

References:

Association	parentFihoXyz	StructureObject	1..1	<i>Equipment</i>
Association	child	Equipment	0..*	<i>StructureObject</i>

9.3 Physical AIS

Type: FeatureAssociationType

Definition:

CamelCase: PhysicalAIS

References:

Association	physicalAISbroadcast By	PhysicalAISaidToNavigation	0..1	<i>RadioStation</i>
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Association	physicalAISbroadcasts	RadioStation	0..*	<i>PhysicalAISaidToNavigation</i>
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9.4 Synthetic AIS

Type: FeatureAssociationType

Definition:

CamelCase: SyntheticAIS

References:

Association	syntheticAISbroadcastBy	SyntheticAISaidToNavigation	0..1	<i>RadioStation</i>
Association	syntheticAISbroadcasts	RadioStation	0..*	<i>SyntheticAISaidToNavigation</i>

9.5 Virtual AIS

Type: FeatureAssociationType

Definition:

CamelCase: VirtualAIS

References:

Association	virtualAISbroadcasts	RadioStation	0..*	<i>VirtualAISaidToNavigation</i>
Association	virtualAISbroadcastBy	VirtualAISaidToNavigation	0..1	<i>RadioStation</i>

9.6 Aton Aggregations

Type: FeatureAssociationType

Definition:

CamelCase: AtonAggregations

References:

Association	peerAtonAggregation	AtonAggregation	0..*	<i>AidsToNavigation</i>
Association	atonAggregationBy	AidsToNavigation	0..*	<i>AtonAggregation</i>

9.7 Aton Associations

Type: FeatureAssociationType

Definition:

CamelCase: AtonAssociations

References:

Association	peerAtonAssociation	AtonAssociation	0..*	<i>AidsToNavigation</i>
Association	atonAssociationBy	AidsToNavigation	0..*	<i>AtonAssociation</i>

9.8 Range System

Type: FeatureAssociationType

Definition: Navigation system limited in their positioning capability to coastal regions, or those systems limited to making landfall

CamelCase: RangeSystem

References:

Association	navigableTrack	RecommendedTrack	0..*	<i>NavigationLine</i>
Association	navigationLineFihoXyz	NavigationLine	1..*	<i>RecommendedTrack</i>

9.9 Dangerous Feature Association

Type: FeatureAssociationType

Definition:

CamelCase: DangerousFeatureAssociation

References:

Association	markingAton	AtonAssociation	1..*	<i>DangerousFeature</i>
Association	danger	DangerousFeature	0..*	<i>AtonAssociation</i>